

Chapter 1: What Is URSAS?

1.1 What Is URSAS?

URSAS (Universal Rail Safety Alert System) is a public transportation safety platform designed to enhance passenger safety and comfort by providing advance notice of unexpected motion and sudden deceleration that may occur during transit.

Conventional railway safety systems have primarily evolved to ensure the safe operation and control of trains. As a result, the operational safety of railway systems has continued to improve over time.

However, risks still remain inside passenger vehicles. Unexpected motion or sudden deceleration can cause passengers to lose their balance or come into contact with others.

URSAS focuses on a different dimension of safety: safety as experienced by passengers.

URSAS is not a system designed to make trains themselves safer.

It is a system designed to create an environment in which passengers can use public transportation with greater confidence, safety, and comfort.

It is a public transportation safety platform intended to improve the sense of security and comfort of each passenger in their everyday journeys.

URSAS does not seek to change how drivers or operators control their vehicles, nor does it directly control train operation. Instead, it provides passengers with information slightly in advance, giving them time to recognize what is about to happen and prepare accordingly.

Even a few seconds of preparation can help transform anxiety into reassurance and tension into readiness, contributing to a safer and more comfortable public transportation experience.

1.2 Why Is URSAS Needed?

In many public transportation systems, passengers frequently travel while standing.

When vehicles enter stations, travel through curves or track switches, undergo operational adjustments to recover from service delays, or encounter other changes in operating conditions, unexpected motion or sudden deceleration may occur.

In such situations, the greatest difficulty for passengers may not be the motion itself, but the fact that it occurs without warning.

This unexpectedness can cause passengers to lose their balance, experience anxiety, or unintentionally come into contact with others.

In crowded vehicles, the loss of balance of a single standing passenger may also affect nearby passengers, potentially creating a chain reaction through the surrounding area.

URSAS seeks to reduce this element of unexpectedness by providing information slightly in advance, giving passengers time to naturally adjust their posture and take appropriate safety actions.

1.3 The Core Philosophy of URSAS

URSAS does not aim to eliminate vehicle motion.

Nor does it aim to automatically control trains or other public transportation vehicles.

Its goal is:

“To enhance safety and comfort by informing passengers slightly in advance.”

Even when the physical motion is the same, people can respond very differently when they have had time to prepare mentally and physically.

Advance awareness can reduce surprise and anxiety while giving passengers time to adjust their posture, secure their balance, or hold onto available support.

URSAS introduces the value of this preparation time into public transportation and proposes a new approach to passenger-centered safety.

1.4 The Concept of “Safety Experience”

Conventional railway safety has primarily focused on:

“Preventing accidents.”

URSAS builds upon this foundation by introducing an additional dimension of safety:

“Providing an experience in which passengers can travel with confidence, safety, and comfort.”

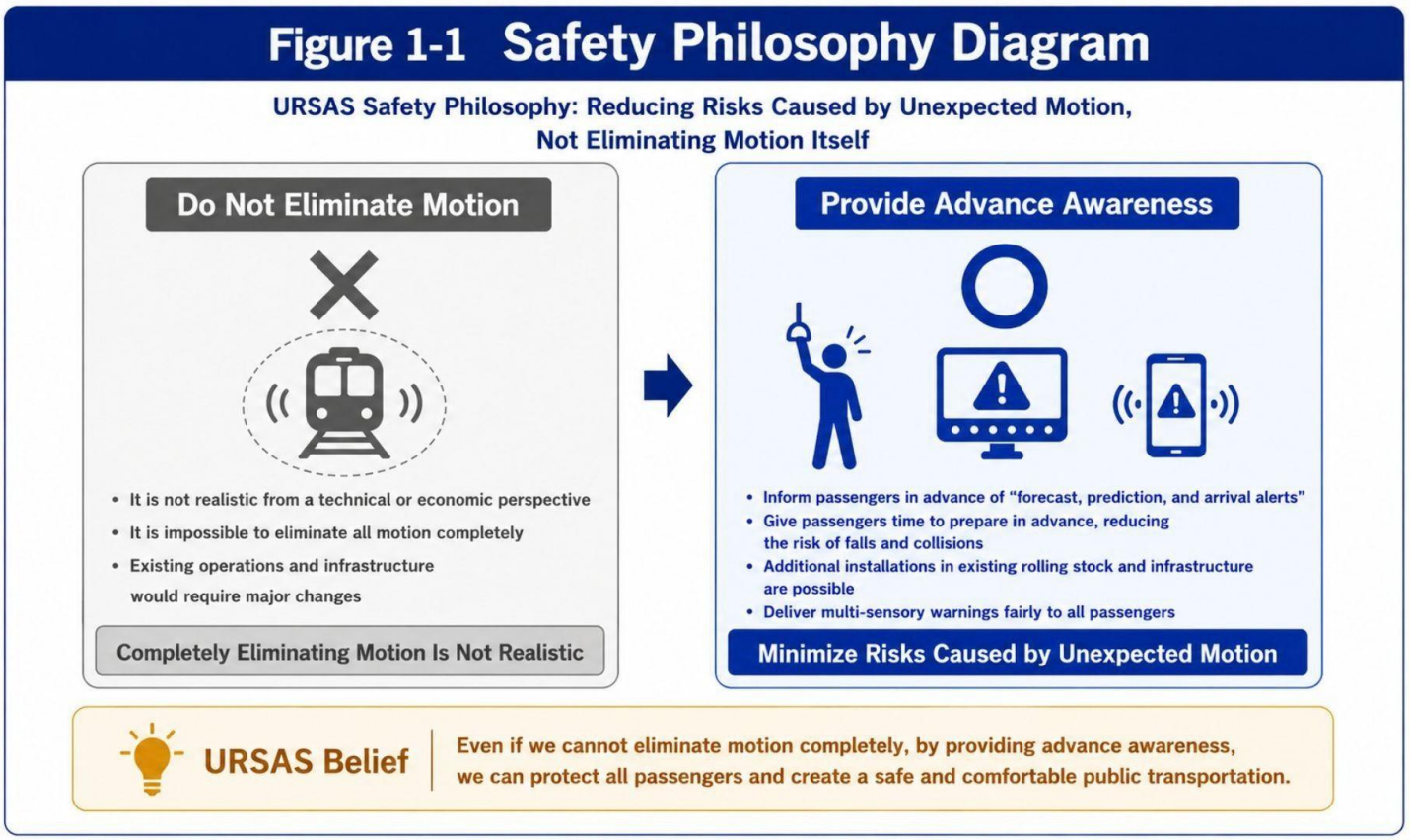
We call this concept:

Safety Experience

Safety Experience goes beyond simply preventing accidents.

By sharing relevant information in advance, URSAS provides passengers with time to prepare, helping them feel more secure and improving their overall experience of public transportation.

URSAS considers safety and comfort not as competing objectives, but as values that can reinforce and enhance one another.



Chapter 2: Public Transportation Experience

2.1 URSAS in 3 Minutes

This section provides an overview of URSAS that can be understood in approximately three minutes. For more detailed technical explanations, please refer to Sections 2.2 and beyond.

URSAS in 3 Minutes

① Why Was URSAS Created?

Background of the Concept

Daily minor discomfort led to the development of URSAS.

After finishing work, I returned to my train at around 5:30 p.m. on a weekday.

During the transfer at the station, I noticed that some passengers stumbled, an elderly person nearly lost their balance, and a child barely kept from falling.

I tried holding the handrail tightly, but the shaking came from a direction I did not expect. Without realizing it, my body leaned, my feet lost stability, and I had to make an unnecessary physical effort just to stay upright.

The vehicle continued running, approaching an area with irregular track conditions. The motion grew larger than before, and I struggled even more to maintain my balance.

I thought to myself, "If only I had known in advance—even by a few seconds—I could have prepared my body and stayed steady." That feeling became the starting point of URSAS.

Although I have experienced many moments of "almost falling," once I started working with rail operators to address these risks, I realized the same feelings were shared by many people. That realization led to the message behind URSAS: "A little earlier awareness can make a big difference."

Insights from This Experience

- ✓ Unexpected motion can happen to anyone.
- ✓ Even a few seconds of advance awareness—if not seconds of preparation—can make a large difference.
- ✓ If the motion experienced in the preceding vehicle can be predicted for the following vehicle, passengers can prepare in advance.

1 After experiencing that moment, I realized that these small daily incidents are not rare at all, but a common experience shared by many passengers.

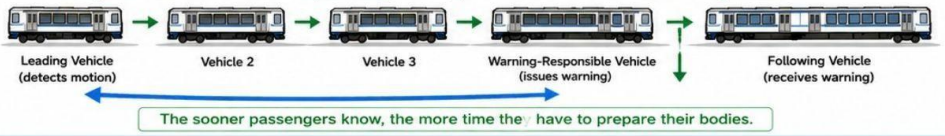


2 This concept aims to provide the kind of awareness and preparation—just a few seconds before unexpected motion occurs—so that passengers can respond in a way that supports their safety and stability.



3 The goal is not to eliminate motion itself, but to provide preparation time for anticipated motion. If passengers have even a few seconds to prepare, they can maintain their posture and reduce risks. Even a small amount of preparation time can make a meaningful difference.

Using the motion detected in the preceding vehicle, predict the motion for the following vehicle.



The sooner passengers know, the more time the have to prepare their bodies.

This subtle awareness can become the starting point of URSAS.

Experience unexpected motion → Can we anticipate motion in advance? → Predict motion earlier → Creating a society where people can ride with confidence

② What Is URSAS, and How Does It Warn Passengers?

URSAS Onboard Advance Warning System – Overall Image

Various devices are integrated to deliver advance awareness and support passengers in making safe and confident movements.

Large Display (Video) — Displays important information such as safety alerts, guidance, announcements, and advertising.

LED Line (Light) — Colored LED lights visually convey alert levels and safety information.

Handrail Vibration Notification — Vibrations in the handrails and stanchions notify passengers by tactile sensations.

Projector (Video) — Projects warning symbols from the ceiling. Animations and colors attract attention and convey alerts.

Speaker (Sound) — Announces alerts and information clearly and appropriately according to the situation.

Aroma Diffuser (Scent) — Provides scents that match the time and situation to create a relaxing atmosphere and deliver advertising content.

Ventilation Unit (Air) — Enhances comfort by adjusting airflow and air quality.

BLE Antenna — Delivers information to smartphones and wearable devices via BLE.



URSAS Operation Flow

Data Acquisition — Sensors · External Data (Train Status · Operation Info)

Three-Layer Prediction (SCE · DPE · DCPE) — Multi-stage risk detection based on route condition · deceleration prediction · observed motion prediction

Tier Core Integration Control — Tier (Reason) + Risk (Intensity) + Kill-Switch for Safety-Priority Control

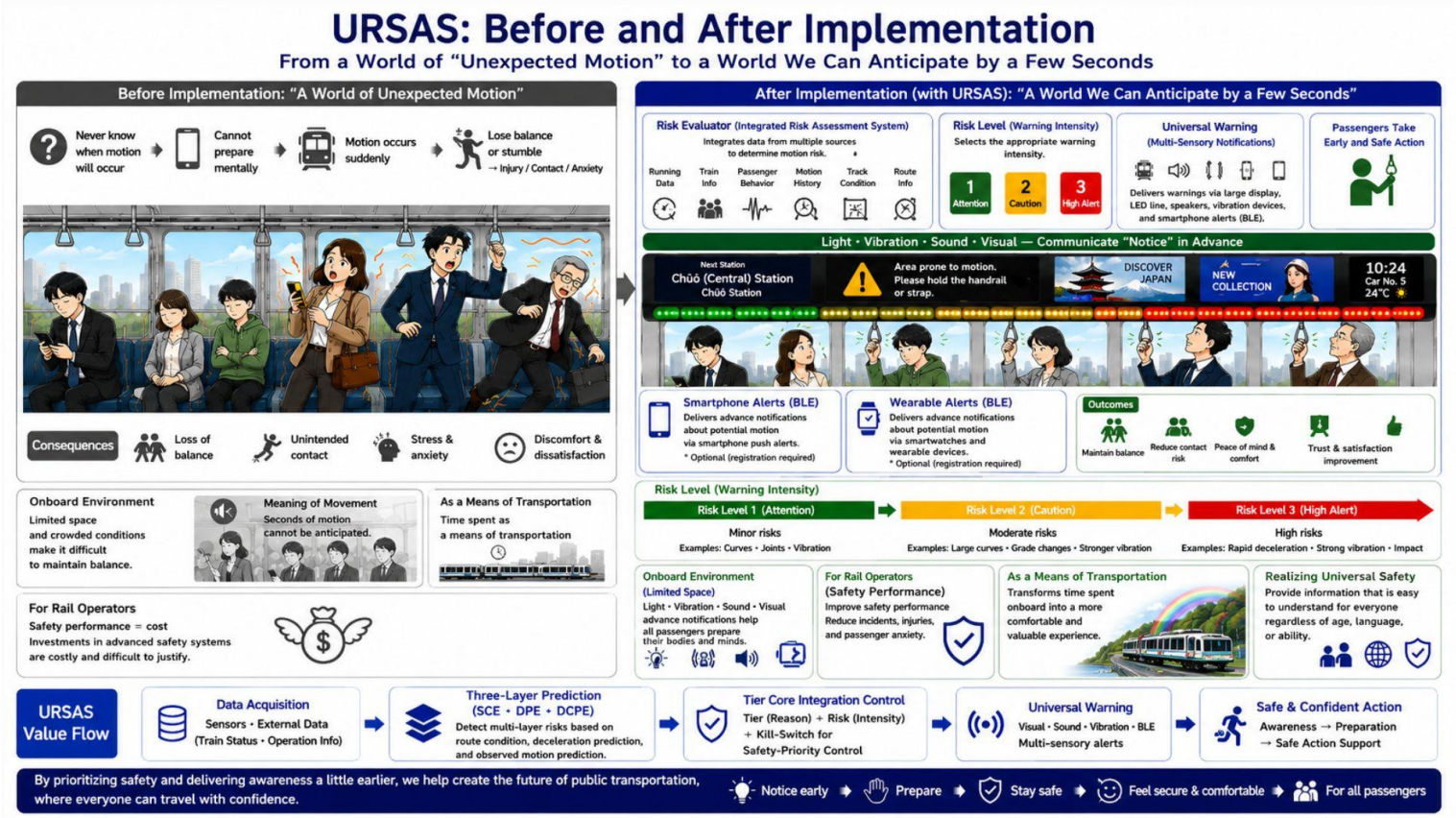
Universal Warning — Visual · Sound · Vibration · BLE Multi-sensory alerts

Safe & Confident Action — Awareness → Preparation → Safe Action Support

By prioritizing safety and providing awareness a little earlier, we create the future of public transportation where everyone can travel with confidence.

Notice early ➡ Prepare ➡ Stay safe ➡ Feel secure & comfortable ➡ For all passengers

③ Before and After



Adding a Human-Centered Public Transportation Experience to Railway Safety Technology

④ Three Prediction and Information-Sharing Technologies

(SCE / DPE / DCPE)

SCE (Tier 1)



Sharing information about locations where motion is more likely to occur

SCE does not assess potential risks based on data from a single train alone.

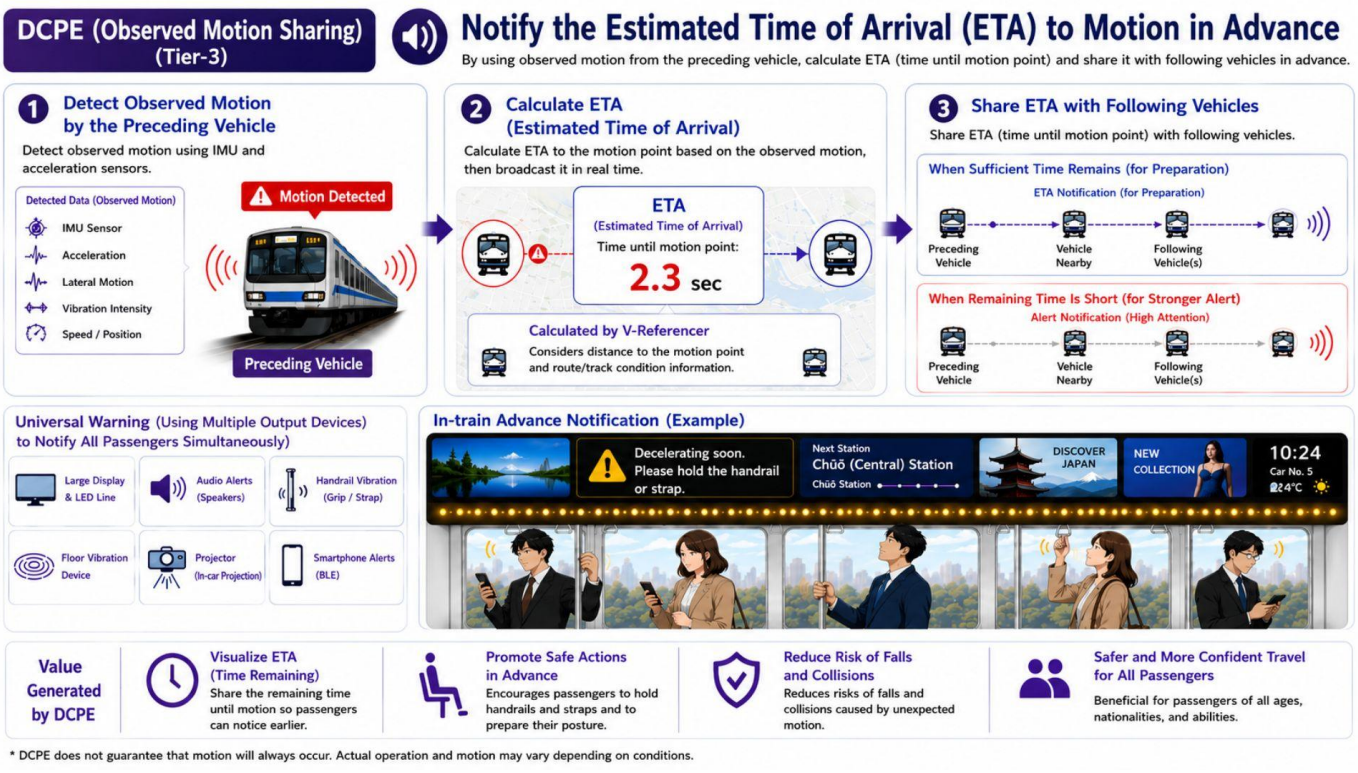
By sharing motion history observed by trains during daily operation, SCE continuously contributes to improving safety across the railway network.

DPE (Tier 2)



Sharing advance information that "deceleration may occur soon"

DCPE (Tier 3)



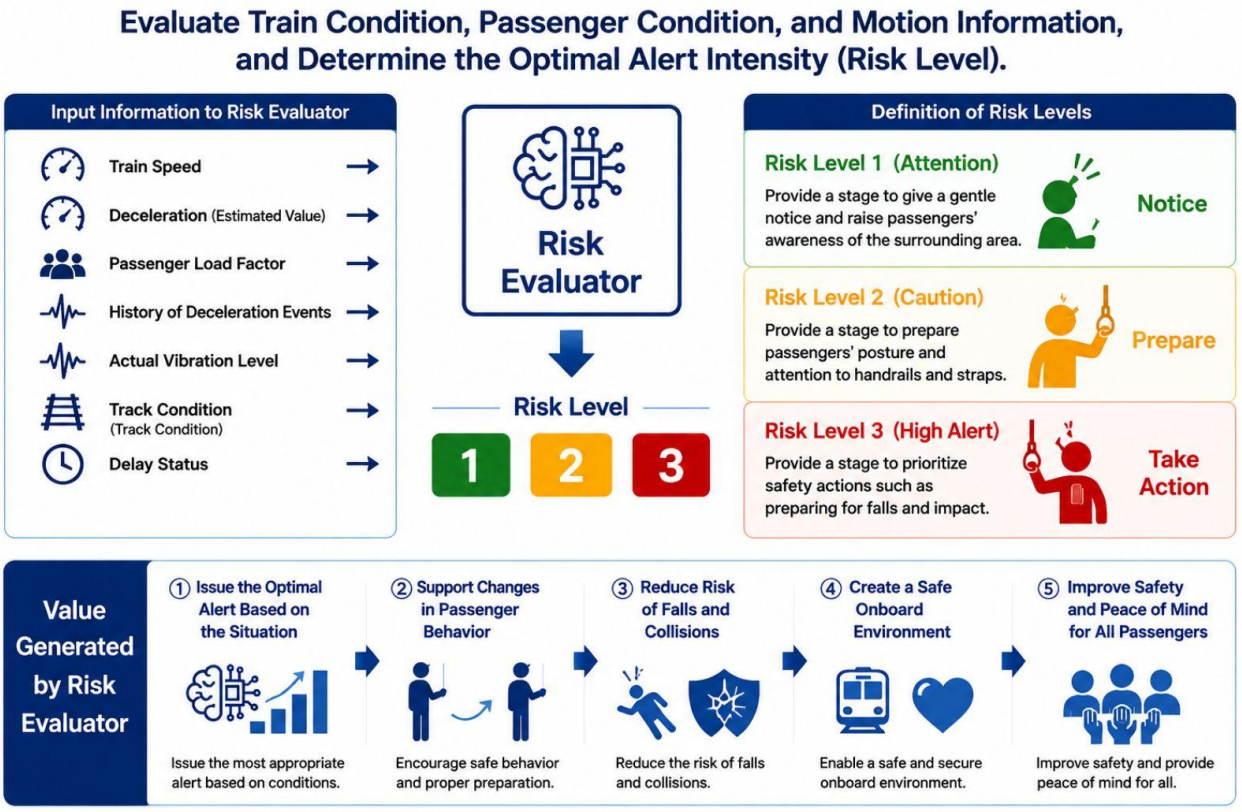
Providing advance notice of the estimated time remaining until the following vehicle reaches the motion point (ETA)

Based on actual Observed Motion detected by a preceding vehicle, DCPE calculates the ETA (Estimated Time of Arrival at the Motion Point) and shares this information in advance with the vehicle immediately behind it.

DCPE uses the measured motion information from a preceding vehicle to calculate the estimated time remaining until the following vehicle reaches the same motion point.

Note: Depending on train speed and vehicle position, sufficient preparation time may not always be available. In such cases, advance information provided by SCE and immediate information sharing from the preceding vehicle complement the warning process.

⑤ Risk Evaluator



A mechanism for determining warning intensity according to the situation

⑥ Intuitive Warning Design

Universal Warning

Using multiple sensory outputs (visual, auditory, tactile), we deliver warnings quickly, directly, and unmistakably to everyone.

Normal (Natural Movement) → Risk1 (Attention) Notice → Risk2 (Caution) Prepare → Risk3 (High Alert) Take Action

Large Display (20 screens)
Communicates train operation information, news, weather, and more, as well as warnings, through text, colors, pictograms, LED lines, and other means.



Normal (Natural Movement)



Risk1 (Attention)



Risk2 (Caution)



Risk3 (High Alert)

	Normal (Natural Movement)	Risk1 (Attention)	Risk2 (Caution)	Risk3 (High Alert)
Audio Alerts (Speakers) Deliver easy-to-understand voice announcements to everyone.	–	Soft chime (1 time)	Medium chime (1 time)	Strong chime (2 times)
Handrail Vibrators (Grip / Strap) Vibrations in handrails and straps let you feel the warning.	–	Light vibration (1 time)	Vibration (1 time)	Strong vibration (Continuous)
Floor Vibration Devices Floor vibrations let you feel the warning with your whole body.	–	Light vibration (1 time)	Vibration (1 time)	Strong vibration (Continuous)
Projector Projection Project visual warnings on floors, walls, or other surfaces.	–	Caution projection	Caution + Direction projection	High alert projection + Direction indication
Smartphone Alerts (BLE) Send alerts to passengers' smartphones via Bluetooth. (Opt-in only)	–	Vibration only	Vibration + Medium alert	Vibration + High alert (Strong)

Purpose of Multi-Modal Notifications: Easy to see (Visual) · Easy to hear (Auditory) · Easy to feel (Tactile) → Deliver immediate and unmistakable safety warnings quickly to everyone.

* Notifications may vary depending on operating conditions, situations, and system settings.

How Universal Warning Works

A multimodal warning system that goes beyond language barriers to deliver messages clearly to all passengers.

Instant slide projection of warning symbols throughout the train



The projector instantly displays numerous warning symbols throughout the train for just one second, enabling passengers to intuitively understand potential dangers.

Attention via smartphone by projector slide projection



Projector slide projections are displayed on smartphone screens, drawing attention even from those not looking up, and delivering the message instantly.

Information conveyed throughout via large displays



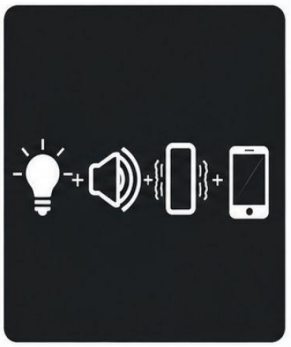
Large displays (approx. 20 m in front-to-back length) deliver information to all passengers at once throughout the entire train.

Vibration delivered through body

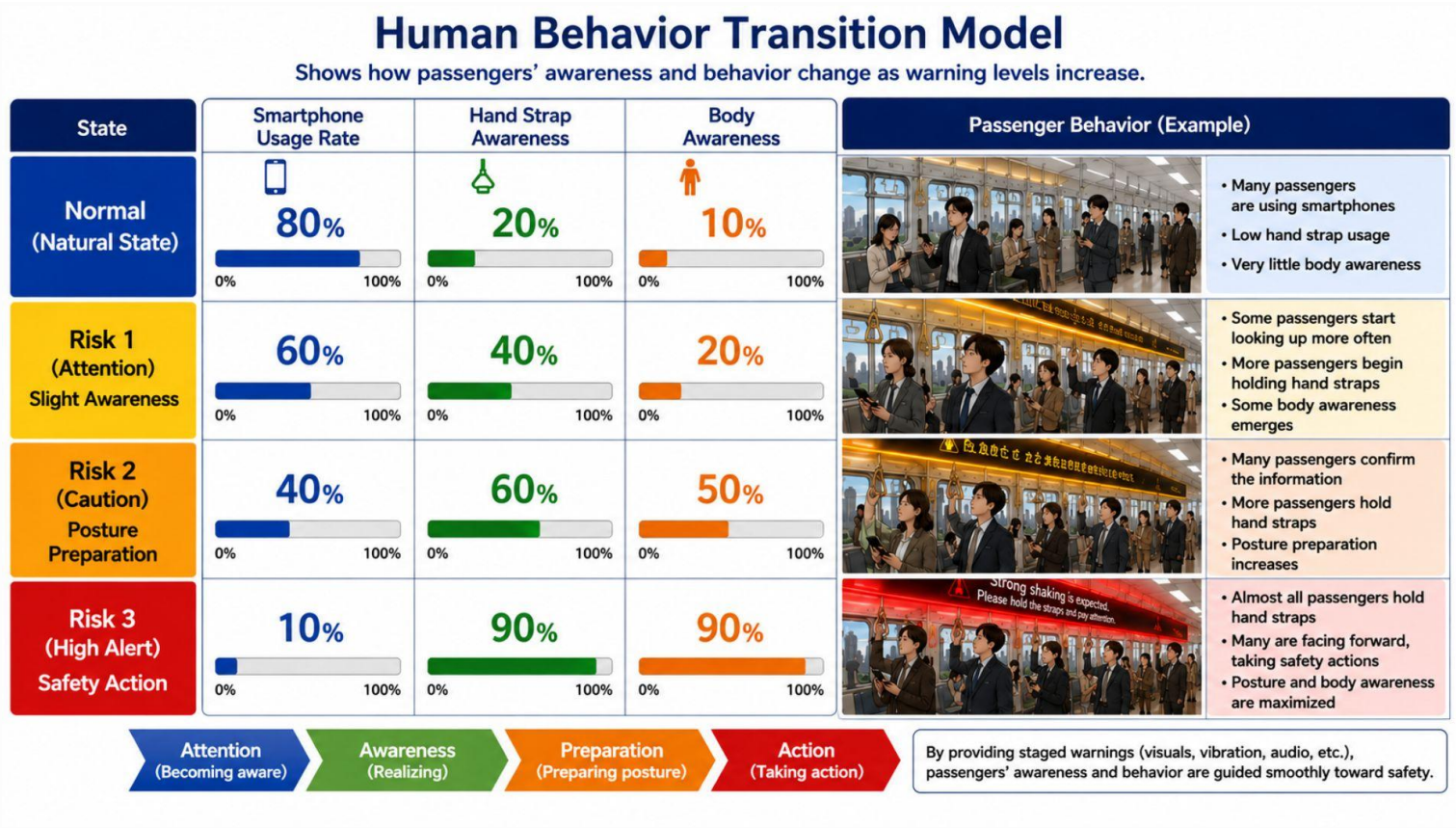


Vibrations of handrails and floor micro-vibrations deliver alerts through both hearing and touch to ensure passengers recognize the danger.

Integrated control of light, video, vibration, and sound

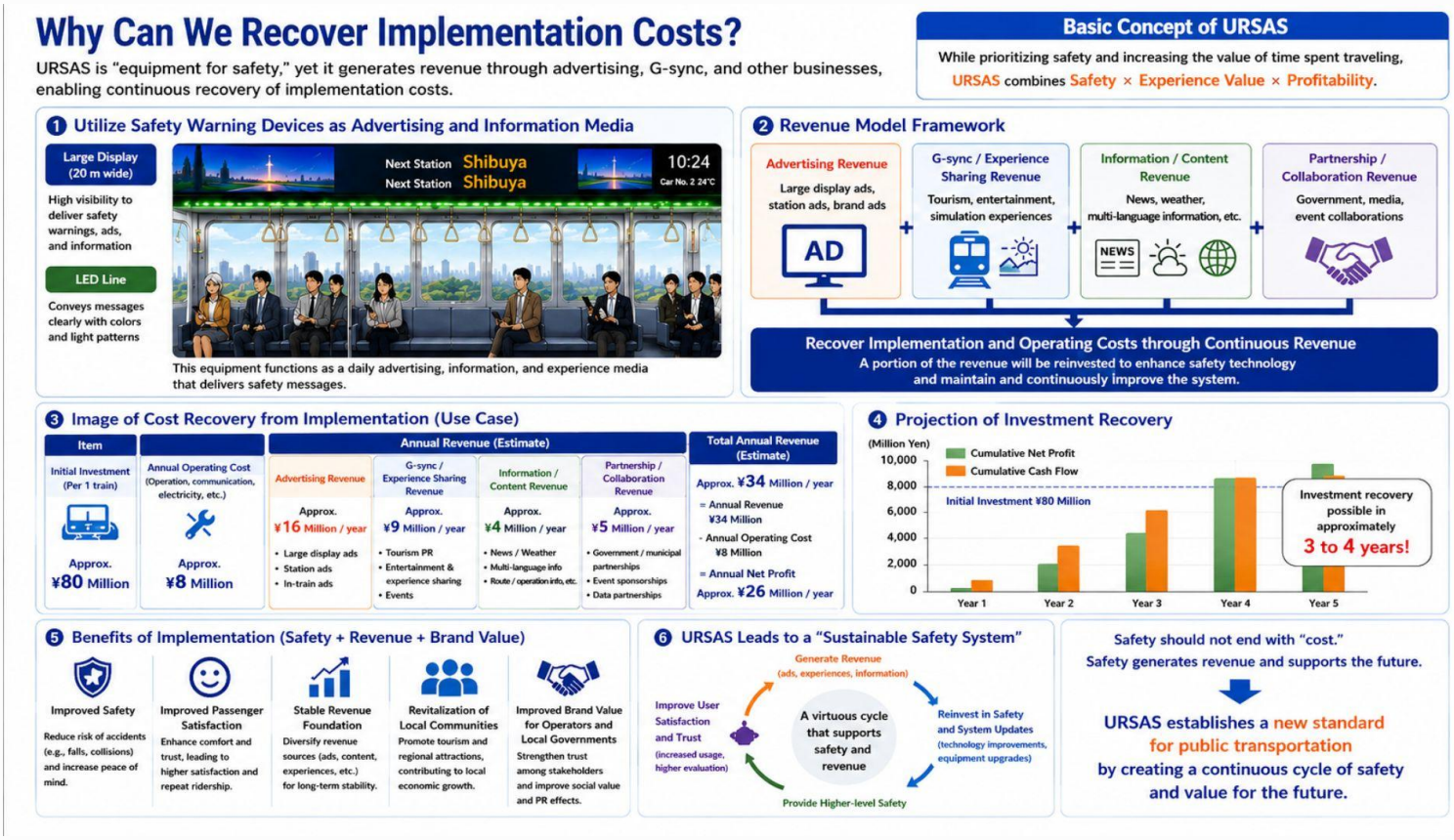


By combining multiple sensory cues, the system delivers warnings more reliably, faster, and more intuitively.



Note: This figure presents a conceptual model of passenger behavior. The percentages shown are illustrative examples only.

⑦ Recovering Implementation Costs



Note: The estimated payback period is based on a model scenario. Actual payback periods may vary depending on the scale of implementation, operating conditions, advertising revenue, sponsorship revenue, and other factors.

⑧ The Future



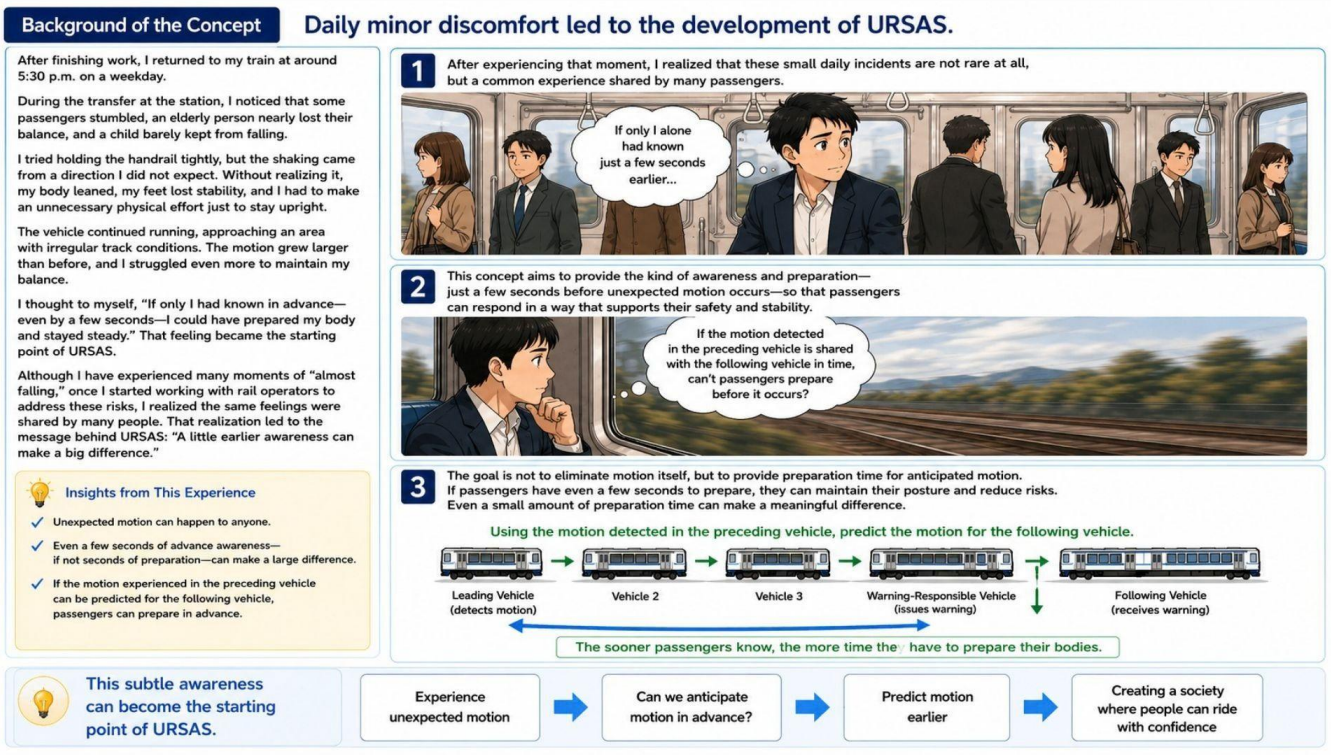
URSAS is more than a technology for warning passengers about motion.

By becoming aware of what is about to happen just a little earlier, people can prepare themselves, support one another, and travel with greater confidence.

Safety, comfort, and passenger experience should not be treated separately. URSAS seeks to bring them together and advance public transportation itself.

By sharing information about unexpected motion in advance, URSAS aims to create a new human-centered public transportation experience.

2.2 Why Was URSAS Created?



While using public transportation in my daily life, I experienced situations in which sudden lateral motion or rapid deceleration caused me to lose my balance significantly.

What I realized from these experiences was:

“The motion itself was not necessarily the greatest danger. The real problem was that I had no time to prepare for it.”

If passengers could know even a few seconds in advance that significant motion was about to occur, many would have time to hold onto a handrail or other support, adjust their posture, and prepare themselves.

URSAS was born from this realization.

2.3 The Problem URSAS Seeks to Address

On public transportation, sudden deceleration or unexpected motion can cause passengers to lose their balance, potentially resulting in falls or contact with other passengers.

Although many of these incidents do not result in serious accidents, they can create everyday safety risks and uncomfortable experiences, such as:

- losing balance, potentially causing discomfort or embarrassment;
- failing to secure a grip on a handrail, strap, or other support;
- unintentionally coming into contact with another passenger, causing concern or fear; and
- experiencing surprise or anxiety.

Even relatively minor motion can place a significant burden on certain passengers, particularly:

- older adults;
- children;
- passengers with reduced physical mobility or stability;
- standing passengers in crowded vehicles; and
- passengers using smartphones or other devices.

Existing railway safety systems, however, have primarily been designed to ensure operational safety.

The perspective of:

“creating time for passengers to prepare for motion”

has not traditionally been their primary focus.

URSAS seeks to address this issue not by:

“eliminating motion,”

but by:

“creating time for passengers to prepare for motion.”

2.4 Core Philosophy

At the heart of URSAS is the concept of:

Reducing Unexpectedness

People react not only to hazards themselves, but also to events that occur:

“without warning.”

Even when the physical intensity of motion is the same, there can be a significant difference in fall risk and psychological burden between:

- anticipated motion, and
- unanticipated motion.

By sharing information about upcoming motion or deceleration in advance, URSAS creates preparation time that allows passengers to take appropriate safety actions.

URSAS is designed according to the following principles:

1. Safety First
2. Human-Centered Design
3. Minimum-Stimulus Principle
4. Multimodal Notification
5. Integration of Safety and Experience Value

2.5 The Concept of Reducing Unexpectedness

URSAS does not primarily target motion itself.

What URSAS seeks to reduce is:

Unexpectedness

Unexpectedness refers to the overall burden that may arise when motion or deceleration occurs without passengers being able to anticipate it, including:

- surprise;
- loss of balance;
- unintended contact with others; and
- anxiety.

Through progressive information sharing, URSAS supports a behavioral transition:

Awareness

↓

Preparation

↓

Safety Action

The purpose is not simply to warn people of danger.

It is to provide passengers with time to prepare, enabling them to travel with greater confidence and security.

2.6 Three-Layer Prediction and Information-Sharing Technologies

URSAS reduces unexpectedness by combining three different prediction and information-sharing technologies.

SCE (Statistical Complement Engine) uses historical motion data and route characteristics to identify and share information about sections where significant motion is more likely to occur.

DPE (Deceleration Prediction Engine) uses train speed and location information to predict and share the possibility of sudden deceleration or strong braking.

DCPE (Deterministic Chain Prediction Engine) uses actual Observed Motion detected by a preceding vehicle as a reference, calculates the ETA (Estimated Time of Arrival at the Motion Point), and shares this information in advance with the vehicle immediately behind it.

Through this three-layer architecture, URSAS progressively utilizes:

Historical statistical information

↓

Current operational and deceleration indicators

↓

Real-time observed motion data

to provide passengers with time to prepare for upcoming motion or deceleration.

2.7 Universal Warning — Notification Distribution Function

Universal Warning does not determine the warning itself.

Its role is to distribute the notification content determined by the Tier Core to the appropriate notification devices according to passenger conditions and the operating environment.

URSAS therefore uses multimodal notifications combining:

- large displays and LED lines;
- audible notifications;
- vibration-based notifications;
- projector displays; and
- smartphone notifications.

By combining multiple communication channels, URSAS is designed to make essential information accessible to passengers regardless of:

- age;
- nationality;
- disability; or
- usage environment.

The objective is to enable passengers with different needs and circumstances to receive the same essential safety information through appropriate sensory channels.

2.8 Public Transportation Experience

URSAS is not designed to pursue safety alone.

During normal operation, it seeks to enhance the value of travel time through experience-oriented functions such as:

- G-Sync (an immersive experience system);
- tourism and destination information;
- multilingual information services; and
- environmental and spatial effects.

These experience functions are provided through a shared platform that includes large onboard displays, onboard networks, and related infrastructure.

The same advance-warning platform can also support revenue-generating services such as advertising, information distribution, and sponsored content, helping offset implementation and operating costs.

Whenever a safety notification is required, however, the Tier Core gives safety notifications priority over experience functions and revenue-generating services.

URSAS aims to provide a public transportation platform that brings together:

Safety × Experience Value × Sustainable Operation

When a safety-related risk occurs, safety notifications always take priority.

URSAS therefore seeks to achieve both:

Safety and Experience Value

without compromising the priority of safety.

2.9 The Future URSAS Aims to Create

The future envisioned by URSAS is not:

“public transportation that warns people of danger.”

It is:

“public transportation in which people can travel with confidence and security.”

If people can become aware of upcoming motion just a little earlier, they can prepare themselves, support one another, and travel with greater confidence.

By sharing information about unexpected motion in advance, URSAS aims to establish a new standard for public transportation in which:

Safety

Comfort

Experience Value

work together in harmony.

Chapter 3: Overall System Architecture

3.1 Overall System Architecture

URSAS is a public transportation support platform that integrates multiple prediction technologies, warning functions, and passenger-experience functions.

The system operates through the following process:

1. Acquire information
2. Predict risks
3. Determine warning content
4. Notify passengers
5. Provide public transportation experiences

At the core of URSAS is a three-layer prediction architecture consisting of:

- SCE (Statistical Complement Engine)
- DPE (Deceleration Prediction Engine)
- DCPE (Deterministic Chain Prediction Engine)

The prediction results generated by these three technologies are integrated through the:

- Tier Engine
- Risk Evaluator
- Tier Core

to determine the final warning policy.

The resulting notification is then delivered to passengers through Universal Warning.

Universal Warning distributes the notification content determined by the Tier Core to various passenger-facing devices, including:

- large displays;
- LED lines;
- audible notification systems;
- vibration-based notification devices;
- projectors; and
- smartphone notifications.

During normal operation, Public Transportation Experience provides functions intended to enhance the value of the journey, including:

- G-Sync, an immersive experience system;
- tourism and destination information;
- multilingual information services; and
- environmental and spatial effects.

The same platform can also support revenue-generating services such as advertising, information distribution, and sponsored content, helping support the continuous operation and maintenance of safety-related infrastructure.

URSAS is therefore designed not simply as a warning system, but as a public transportation support platform that integrates:

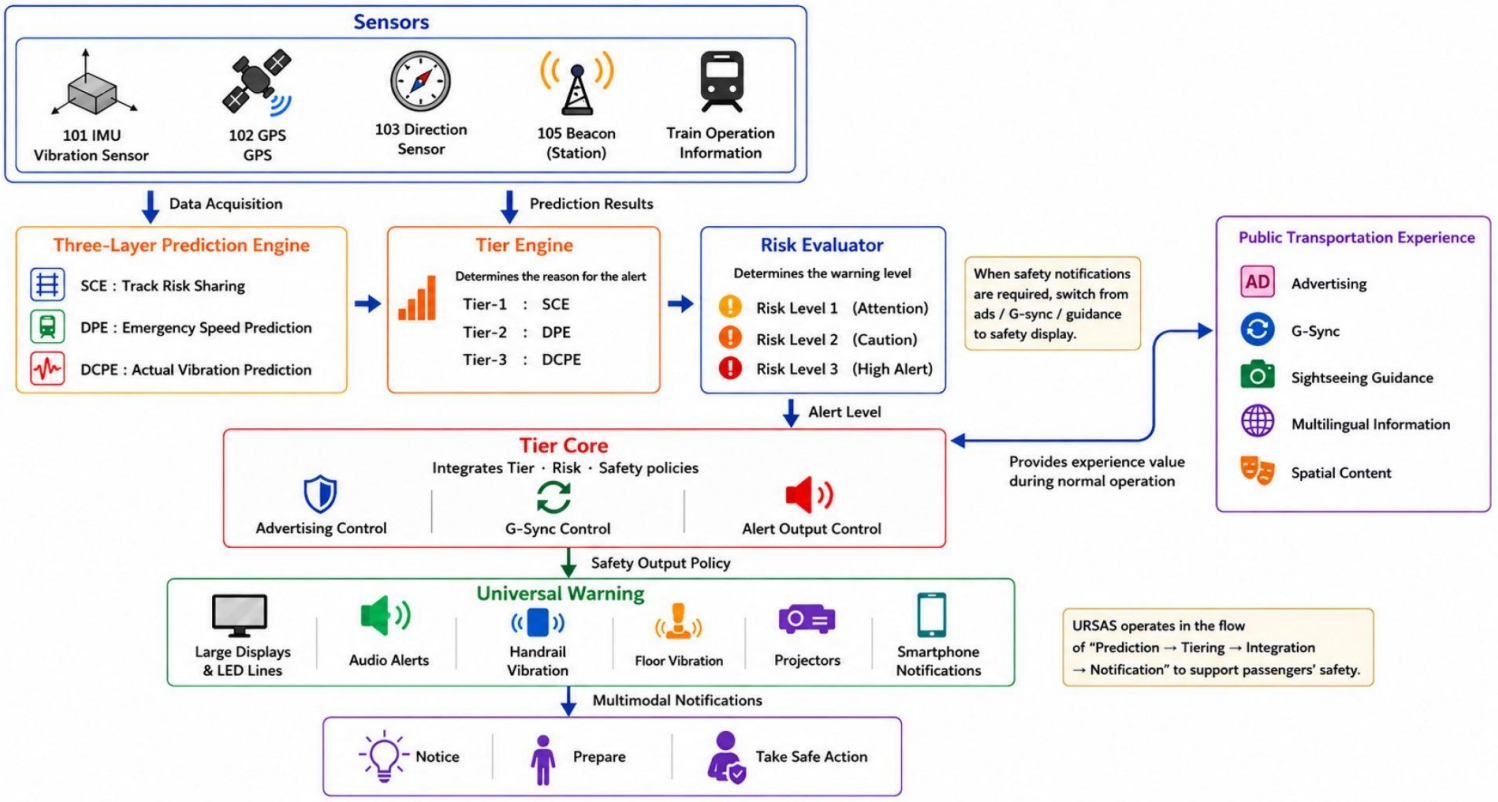
Safety

and

Experience Value

within a single system architecture.

Fig. 3-1 URSAS 5.4 System Architecture (Overview)



3.2 Function Numbering System

This document uses a function numbering system to make each URSAS function easier to identify and manage.

100 Series: Sensors

Functions that acquire information about vehicle conditions and the external environment.

Examples:

101 IMU

102 GPS

103 Direction Sensor

104 Audio Detection

105 Station Beacon Receiver

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200 Series: Control & Management

Functions responsible for managing and coordinating the overall system.

Examples:

201 Main MCU

202 Safety MCU

203 Effect MCU

204 Role Manager

205 Redundancy Manager

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300 Series: Prediction & Analytics

Functions used to predict and analyze motion and deceleration.

Examples:

301 DCPE

302 SCE

303 DPE

304 V-Referencer

305 ETA Calculation

306 Risk Evaluator

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400 Series: Warning Outputs

Functions used to communicate safety information to passengers.

Examples:

401 Large Display & LED Line

402 Projector

403 Speaker

404 Handrail / Support-Bar Vibrator

405 Floor Vibration Device

408 Smartphone & Wearable Notification

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500 Series: Governance

Functions that implement safety-priority control and system governance.

Examples:

501 Tier Core

502 Kill-Switch

503 Safety Policy

504 Advertising Priority Control

505 EEW Integration

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600 Series: Public Transportation Experience

Functions designed to enhance the value of the passenger journey and overall transportation experience.

Examples:

601 G-Sync

602 Station Geofence

603 Aroma Control

604 Airflow Control

605 Tourism Information

606 Interactive Advertising

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700 Series: External Integration

Functions that interface with external systems and services.

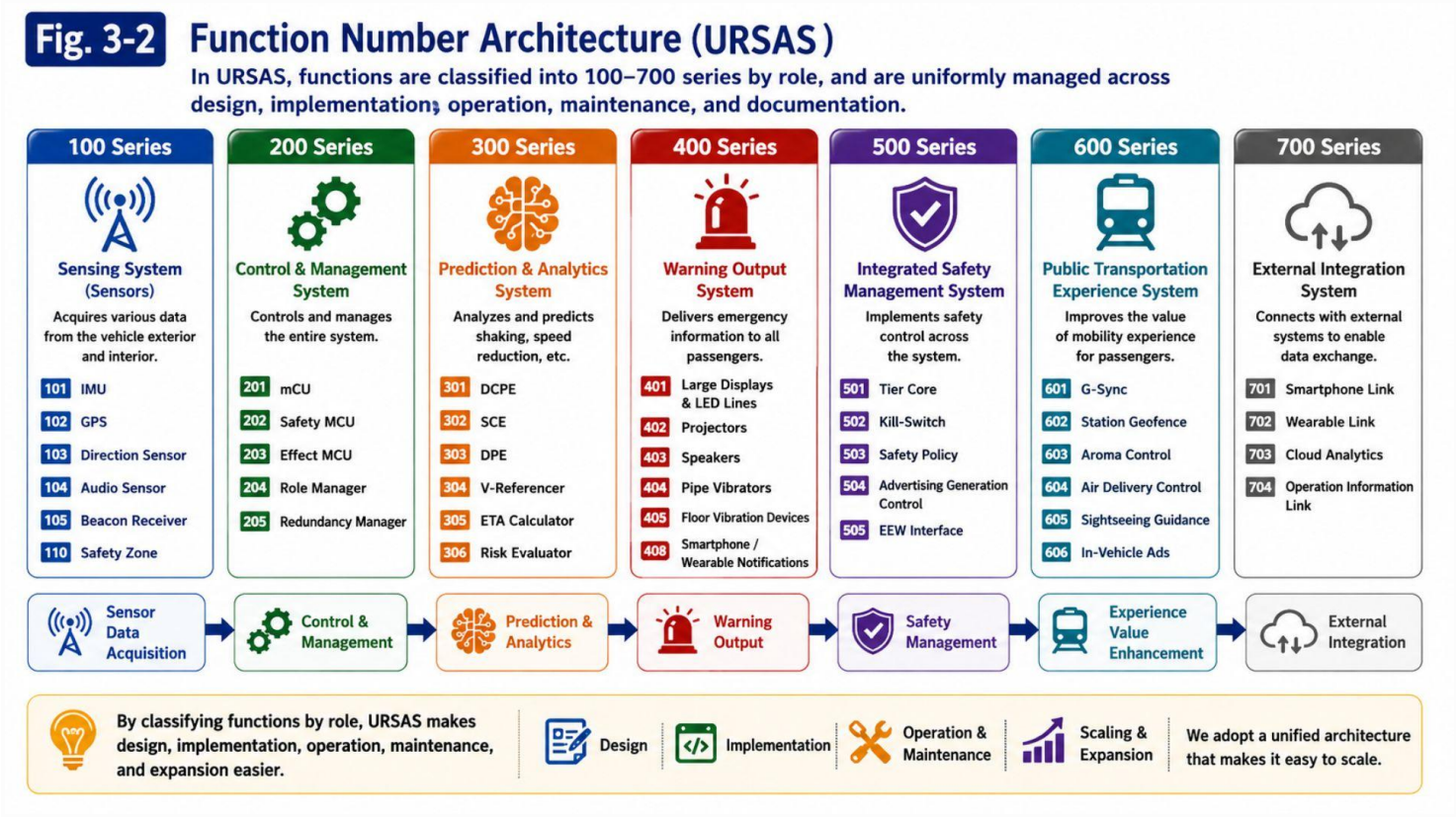
Examples:

701 Smartphone Integration

702 Wearable Integration

703 Cloud Analytics

704 Operational Information Integration



3.3 Functional Layers

From an operational perspective, URSAS is organized into six functional layers.

These operational layers describe how the system functions from information acquisition through prediction, decision-making, passenger notification, governance, and experience delivery. They are conceptually distinct from the seven function-number series defined in the Function Numbering System, which are used to classify and manage individual URSAS functions.

The six functional layers are as follows:

1. Detection Layer
Acquires information required for system operation through onboard sensors and external information sources. This includes vehicle motion, location, speed, direction, track-related information, and other operational data.

2. Prediction Layer
Analyzes acquired information through the three prediction engines—SCE, DPE, and DCPE—to identify potential motion-related events and provide advance information.

3. Decision Layer
Evaluates prediction-engine outputs and current operating conditions. The Tier Engine identifies the warning reason, while the Risk Evaluator determines the appropriate warning intensity. The Tier Core integrates these results and determines the final output policy.

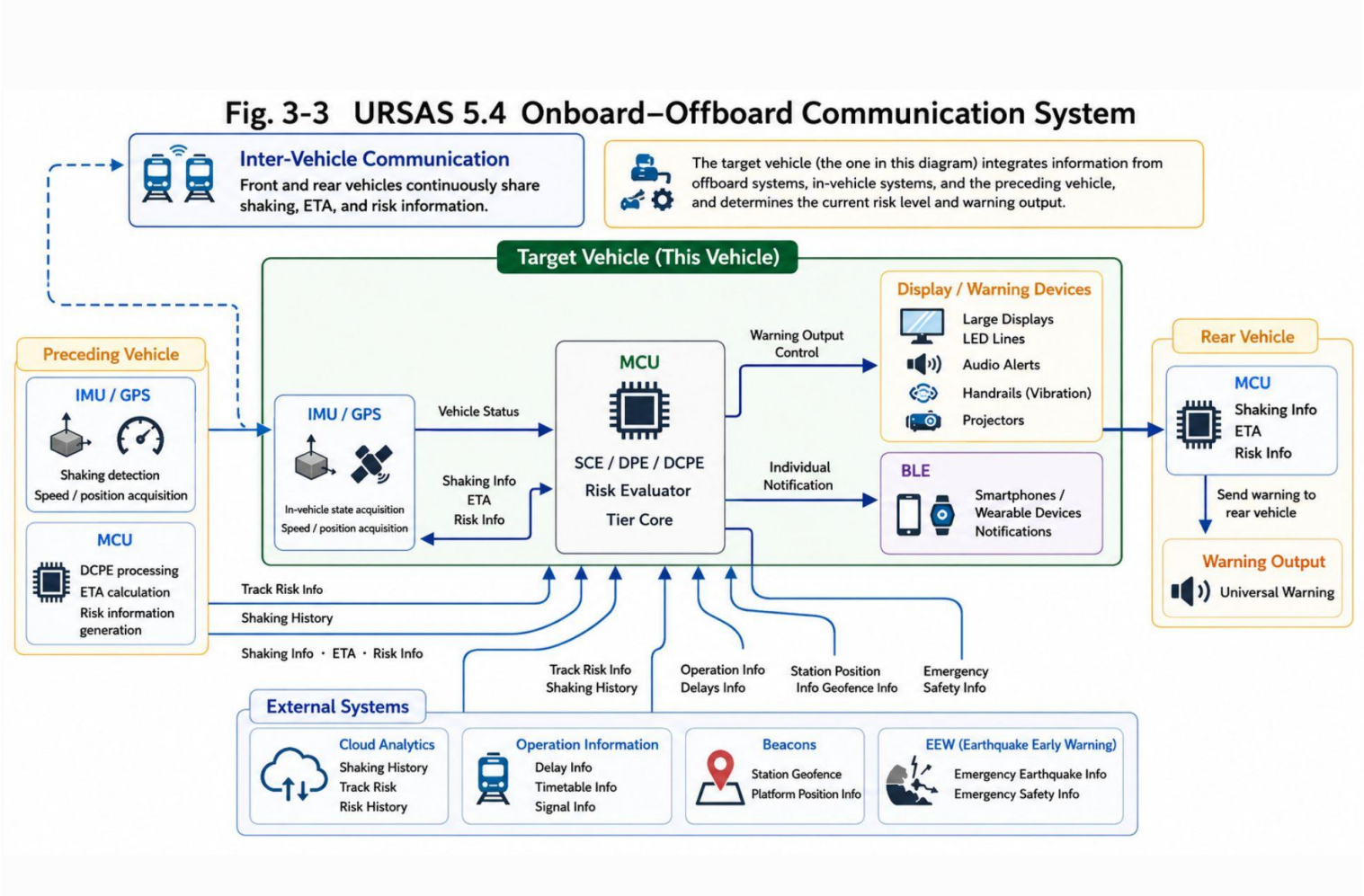
4. Notification Layer
Communicates safety information to passengers through Universal Warning using multiple output devices, including large onboard displays, LED lines, audio alerts, vibration devices, projectors, and smartphone notifications.

5. Governance Layer
Maintains system-wide safety priority and control through mechanisms such as the Safety-First Principle, Kill-Switch, fail-safe design, and integration with Earthquake Early Warning (EEW).

6. Experience Layer
Provides passenger-experience functions during normal operation, including advertising, G-Sync, sightseeing guidance, multilingual information, and other experiential content. These functions always yield priority to safety-related functions when a safety notification is required.

Together, these six operational layers describe the functional flow of URSAS from information acquisition to passenger-facing output.

The seven function-number series, by contrast, provide a separate architectural classification for identifying and managing individual system functions. Therefore, the six functional layers and the seven function-number series represent two complementary views of the same URSAS architecture rather than conflicting system structures.



Chapter 4: Tier-Based Safety Experience Control

4.1 Overview of Tier-Based Safety Experience Control

URSAS is not simply a warning system.

The system supports passenger safety behavior by using separate functions to determine:

- why a warning should be issued;
- how strongly passengers should be warned; and
- how safety should be secured in emergency situations,

and then integrating these functions.

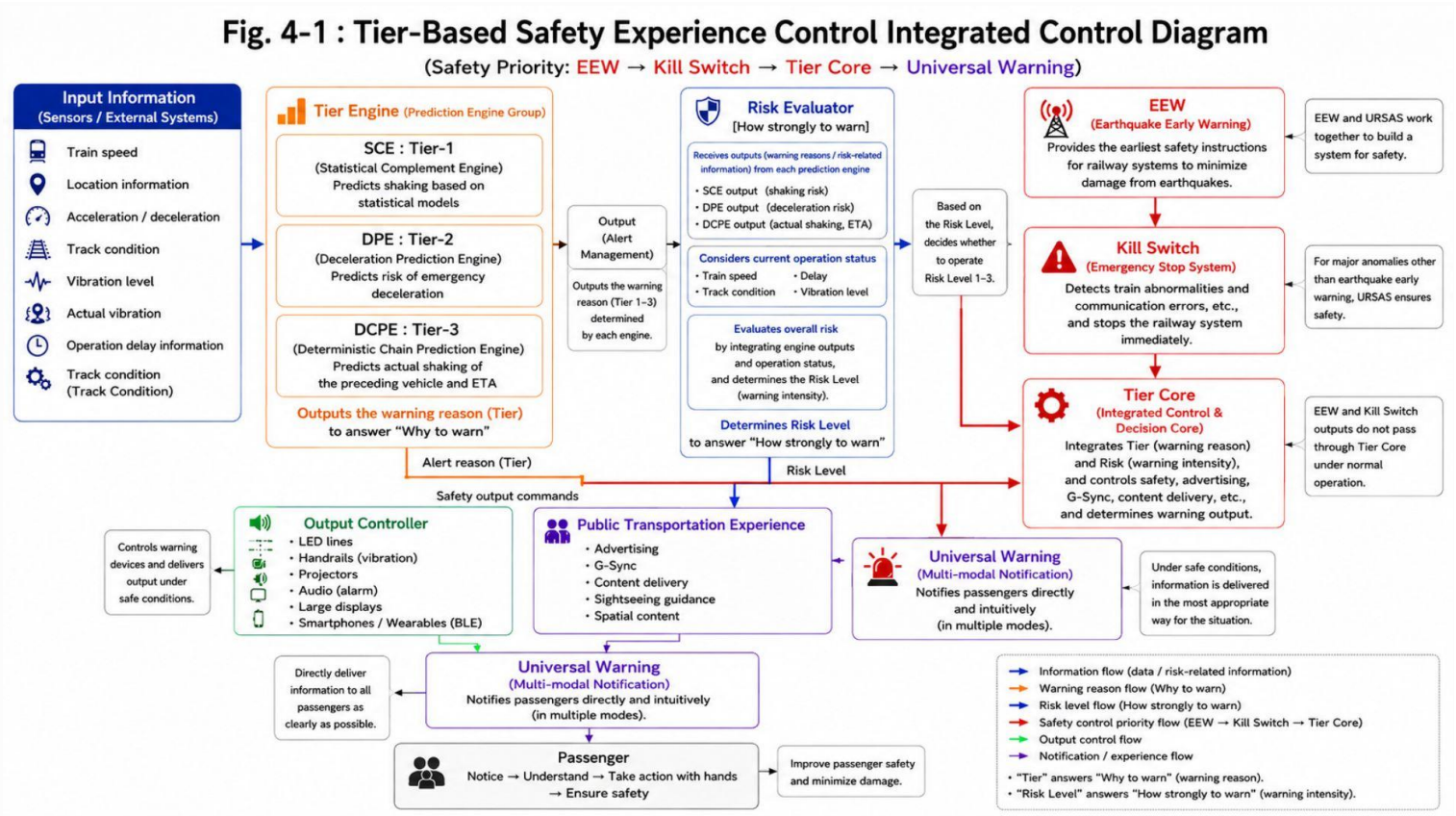
The purpose of URSAS is not to eliminate vehicle motion itself, but to:

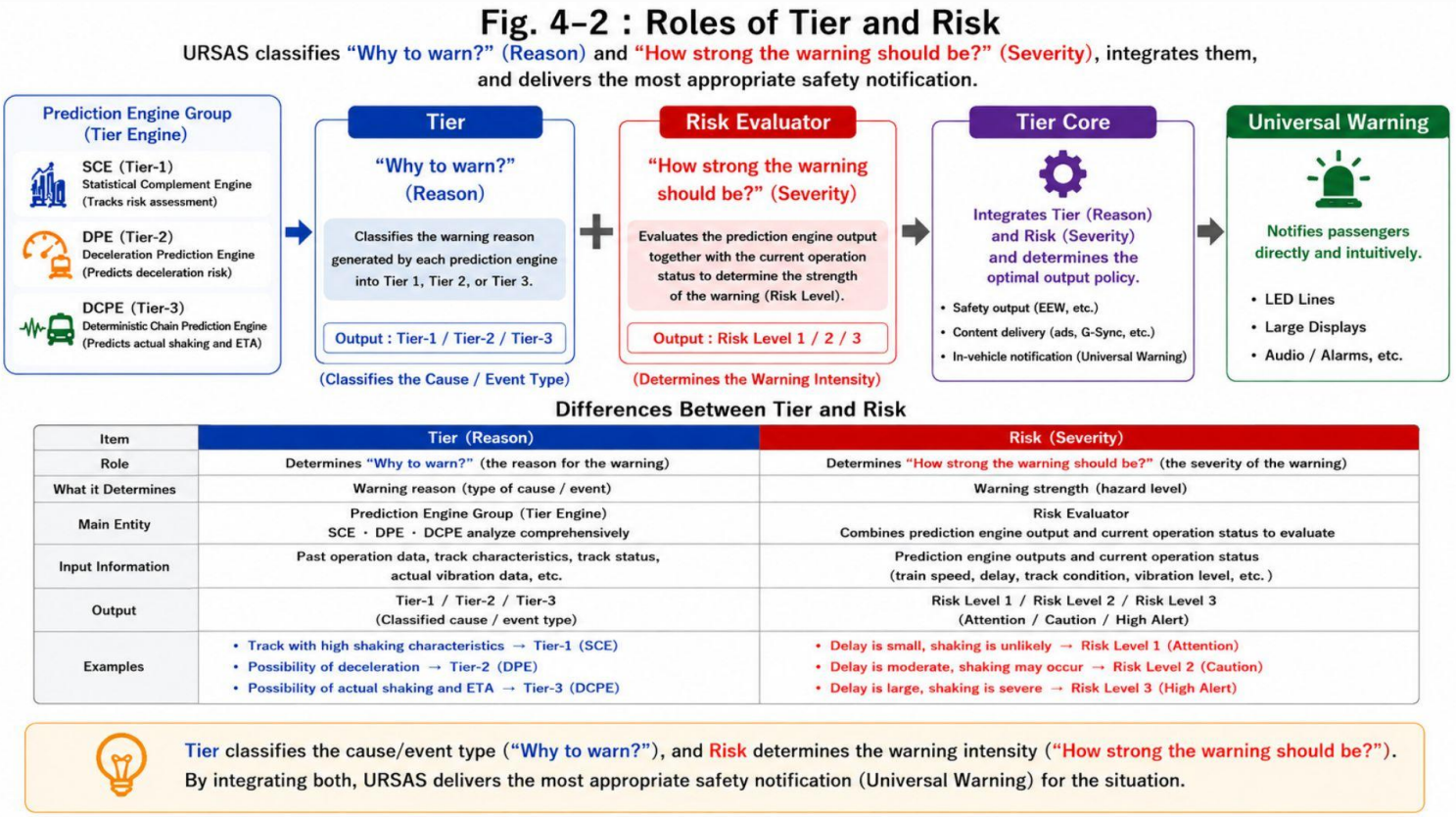
“create time for passengers to prepare for motion or sudden deceleration.”

For this reason, URSAS appropriately combines the reason for a warning with its warning intensity according to the situation, providing warnings that support passenger safety behavior without unnecessarily startling passengers.

This concept of progressive safety and experience control is referred to as:

Tier-Based Safety Experience Control





Differences Between Tier and Risk

Item	Tier (Reason)	Risk (Severity)
Role	Determines "Why to warn?" (the reason for the warning)	Determines "How strong the warning should be?" (the severity of the warning)
What it Determines	Warning reason (type of cause / event)	Warning strength (hazard level)
Main Entity	Prediction Engine Group (Tier Engine) SCE · DPE · DCPE analyze comprehensively	Risk Evaluator Combines prediction engine output and current operation status to evaluate
Input Information	Past operation data, track characteristics, track status, actual vibration data, etc.	Prediction engine outputs and current operation status (train speed, delay, track condition, vibration level, etc.)
Output	Tier-1 / Tier-2 / Tier-3 (Classified cause / event type)	Risk Level 1 / Risk Level 2 / Risk Level 3 (Attention / Caution / High Alert)
Examples	• Track with high shaking characteristics → Tier-1 (SCE) • Possibility of deceleration → Tier-2 (DPE) • Possibility of actual shaking and ETA → Tier-3 (DCPE)	• Delay is small, shaking is unlikely → Risk Level 1 (Attention) • Delay is moderate, shaking may occur → Risk Level 2 (Caution) • Delay is large, shaking is severe → Risk Level 3 (High Alert)

Tier classifies the cause/event type ("Why to warn?"), and Risk determines the warning intensity ("How strong the warning should be?").
By integrating both, URSAS delivers the most appropriate safety notification (Universal Warning) for the situation.

4.2 Mechanism for Determining the Reason for a Warning — Tier Engine

The Tier Engine is the function that determines:

"Why to Warn"

— the reason for issuing a warning.

URSAS classifies warning reasons based on information obtained from its three-layer architecture:

- **Tier 1: SCE (Statistical Complement Engine)**
- **Tier 2: DPE (Deceleration Prediction Engine)**
- **Tier 3: DCPE (Deterministic Chain Prediction Engine)**

The primary warning reasons handled by each Tier are as follows.

Note: Each cell in the Tier × Risk structure represents an example combination of Tier information and Risk Level. The individual prediction engines do not directly determine the Risk Level. The Risk Evaluator determines the Risk Level by integrating multiple sources of information.

Tier 1 — SCE: Sharing Route-Risk Information

Based on risk-related information generated from historical records of Observed Motion and route characteristics, SCE provides a reason for issuing advance warnings for sections that may require increased passenger awareness.

Tier 2 — DPE: Sharing Sudden-Deceleration Risk

Based on vehicle speed, position, operating conditions, historical observation data, and other relevant information, DPE predicts the possibility of sudden deceleration and provides the prediction result as a reason for issuing a warning.

Tier 3 — DCPE: Sharing Actual Observed Motion

DCPE uses Observed Motion actually detected by a preceding vehicle and calculates the ETA until a following vehicle reaches the same Motion Point, thereby providing a reason for issuing an advance warning.

Tier does not determine warning intensity.

The role of the Tier Engine is to determine:

“Why to Warn.”

The Risk Level, which represents warning intensity, is determined separately by the Risk Evaluator.

4.3 Central Function for Determining the Final Warning Policy — Tier Core

The Tier Core is the central integrated-control function within URSAS.

The Tier Core integrates:

- the warning reason (Tier) determined by the Tier Engine;
- the warning intensity (Risk) determined by the Risk Evaluator; and
- safety-priority control by the Kill-Switch,

and determines the final warning policy.

In other words:

Tier Engine → Why to Warn
Risk Evaluator → How Strongly to Warn
Tier Core → Final Warning Policy

The Tier Core also manages priorities among:

- advertising control;
- G-Sync control; and
- experience-effect control,

while always giving the highest priority to safety control.

When a safety warning becomes necessary, safety-related information takes priority over advertising and experience-oriented content.

Fig. 4-3 : Tier × Risk Structure Table

Tier = Why to Warn? (Reason) / Risk Level = How Strongly the Warning Should Be? (Severity)

Risk Level \ Tier	Tier-1 (SCE) Route-Risk Information	Tier-2 (DPE) Sudden-Deceleration Risk	Tier-3 (DCPE) Actual Observed Motion
	Based on historical records of Observed Motion and route characteristics, shares information on sections where motion is more likely to occur.	Based on train operating status, delay, station location, etc., predicts the possibility of sudden deceleration or strong braking.	Based on actual Observed Motion detected in the preceding vehicle, predicts and shares the arrival time at the same Motion Point (ETA).
Risk Level 1 (Attention)	Slight motion history range	Normal deceleration (Deceleration level: Low)	Small actual motion (Amplitude: Small)
Risk Level 2 (Caution)	Moderate motion history range	Moderate to strong deceleration (Deceleration level: Medium)	Moderate actual motion (Amplitude: Medium)
Risk Level 3 (High Alert)	Severe motion history range (High probability / Contact risk)	Possibility of emergency deceleration due to delay recovery operation (Deceleration level: High / Reversal risk: High)	Large actual motion (Amplitude: Large / High probability)

Tier shows "Why to Warn?" (Reason), and Risk Level shows "How Strongly the Warning Should Be?" (Severity).
By integrating both, URSAS delivers the most appropriate warning according to the situation.

4.4 Mechanism for Communicating with Passengers — Universal Warning

The final warning policy determined by the Tier Core is communicated to passengers through:

Universal Warning

Universal Warning does not determine the reason for a warning or its warning intensity.

It is the final output mechanism that delivers information to passengers through appropriate output methods based on the warning policy determined by the Tier Core.

Primary output methods include:

- large displays and LED lines;
- audible notifications;

- handrail / support-bar vibrators;
- floor vibration devices;
- projectors; and
- smartphone and wearable notifications.

By combining these methods, URSAS communicates information through:

visual, auditory, and tactile channels

and aims to provide warnings that can be intuitively understood by as many passengers as possible, regardless of age, nationality, or disability.

4.5 Integrating Safety and Experience Value — Public Transportation Experience

URSAS aims not only to improve safety but also to enhance the value of the public transportation experience itself.

During normal operation, functions and content such as:

- G-Sync, an experience-oriented presentation system;
- tourism and destination information;
- multilingual information services;
- advertising on large onboard displays;
- environmental and spatial effects; and
- sponsored content

may be used to enhance the value of travel time itself.

However, whenever a safety-related notification becomes necessary:

Safety always takes priority.

Public Transportation Experience does not compete with safety functions. It is a framework for enhancing the value of the public transportation experience while maintaining safety as its foundation.

4.6 Mechanism for Determining Warning Intensity — Risk Evaluator

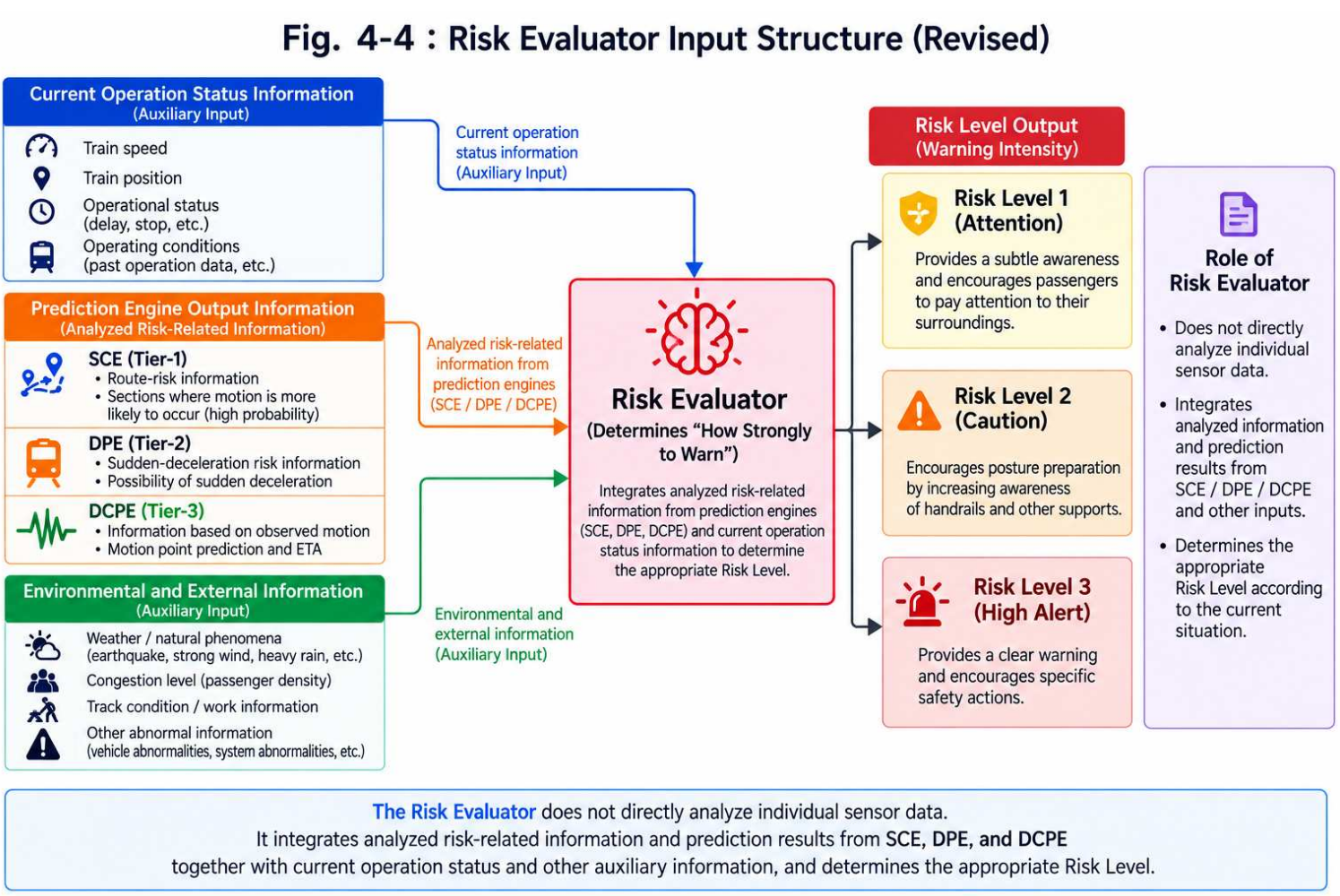
The Risk Evaluator is the function that determines:

“How Strongly to Warn.”

The Risk Evaluator does not determine the Risk Level by directly analyzing individual sensor data.

Instead, it integrates analyzed risk-related information and prediction results output by SCE, DPE, DCPE, and other relevant functions, and determines the appropriate Risk Level according to the current situation.

URSAS uses the following three Risk Levels.



Note: The Risk Evaluator does not directly analyze individual sensor data. Instead, it integrates the outputs of prediction engines such as SCE, DPE, and DCPE and determines the appropriate Risk Level according to the current operational conditions.

Risk Level 1 — Attention

Awareness Phase

Provides passengers with a subtle indication that encourages awareness of their surroundings.

Risk Level 2 — Caution

Posture Preparation Phase

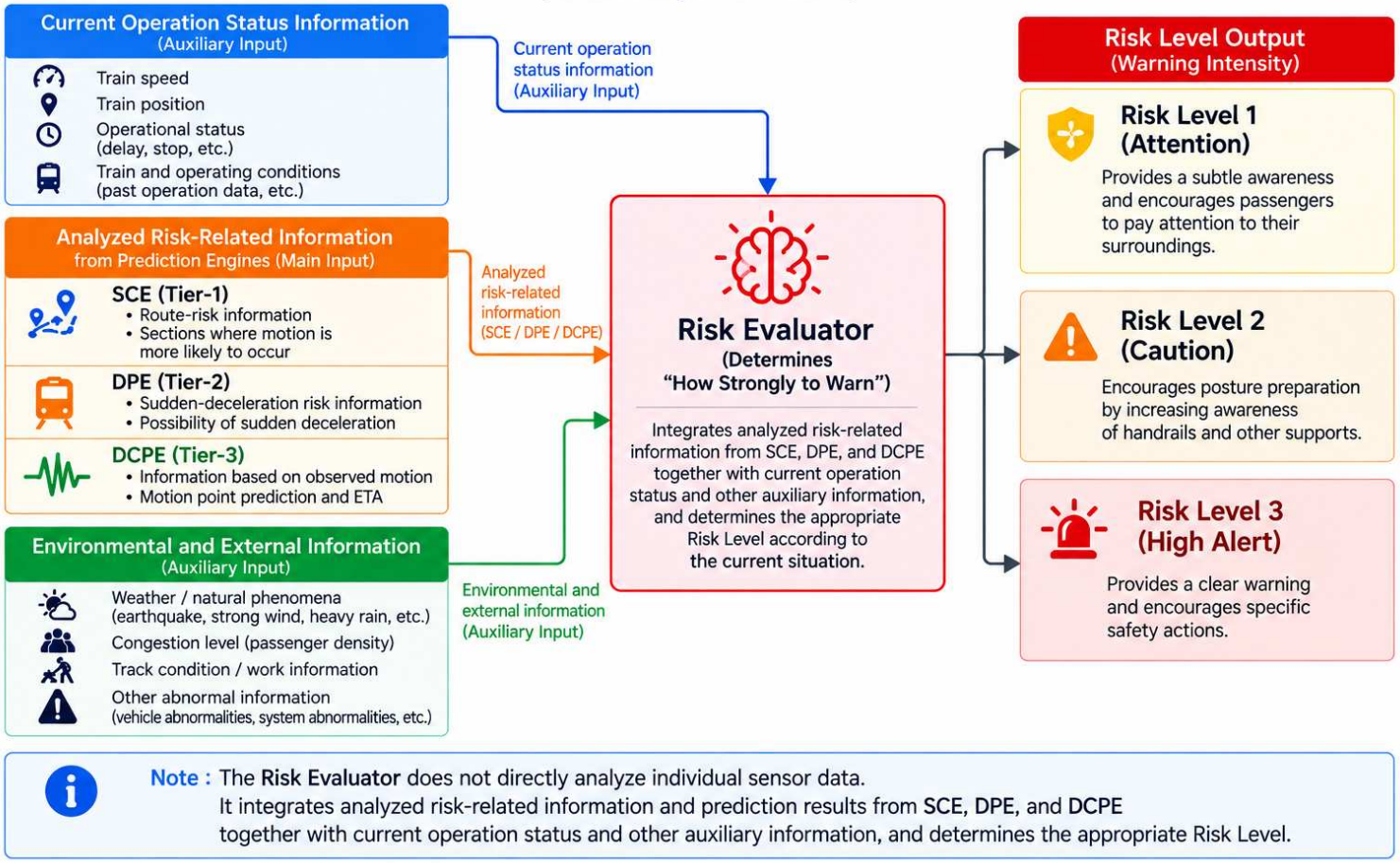
Encourages passengers to become aware of handrails, straps, and other available supports and helps them prepare their posture for motion or sudden deceleration.

Risk Level 3 — High Alert

Safety Action Phase

Provides a clearer warning and encourages passengers to take specific safety actions.

Fig. 4-5 : Information Flow to the Risk Evaluator
(Revised Input Structure)



Therefore, the roles are clearly separated as follows:

Tier Engine = Warning Reason
Risk Evaluator = Warning Intensity

Tier-Based Safety Experience Control

Fig. 4-6 : Information Flow and Role Separation in URSAS
(Revised to Latest Structure)

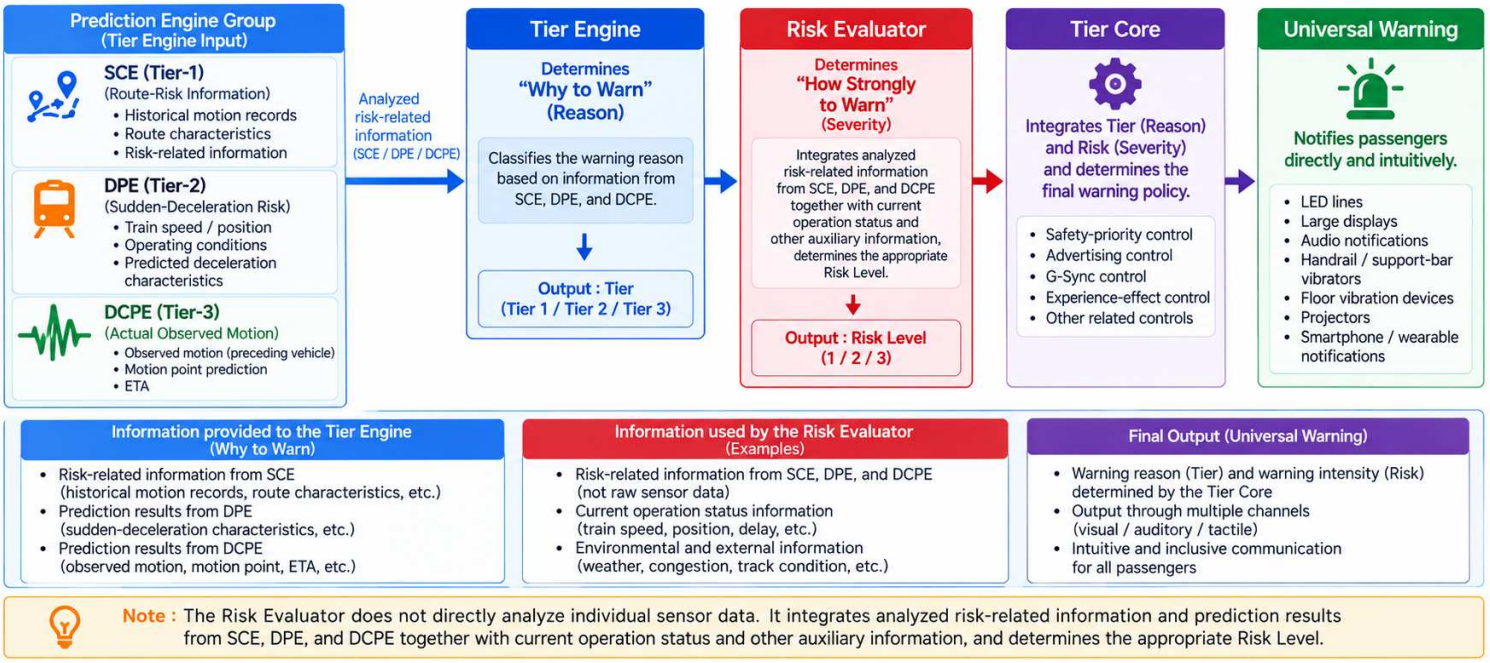


Fig. 4-7 : Output Devices by Risk Level

As the Risk Level increases, the types and intensity of output devices are enhanced in stages.
Provide sufficient attention alerts as needed to promote passenger safety actions.

Device	Risk 1 (Attention)	Risk 2 (Caution)	Risk 3 (High Alert)
LED Line	●	●	●
Hand Strap Vibration	●	●	●
Floor Vibration	–	●	●
Projector	–	●	●
Audio Notification	–	1 time	2 times
Large Display	–	●	●
Smartphone Notification	–	●	●
Advertising / Announcements	Continuous	Continuous	Stop
G-Sync	Continuous	Stop	Stop

● Weak (Notification) ● Medium (Warning) ● Strong (High Alert) – Not used Continuous Continuous possible Stop Stopped

---

4.7 Final Safety Mechanism — Kill-Switch

The Kill-Switch is the final internal safety mechanism within URSAS and operates at a higher priority than normal Tier-Based Safety Experience Control.

The Kill-Switch is activated when a serious event is detected for which normal control should not continue, including coordination with:

- EEW (Earthquake Early Warning);
- vehicle abnormalities;
- communication failures; and
- system failures.

When the Kill-Switch is activated, safety control takes priority over normal advertising, G-Sync, experience-oriented effects, and other functions.

When necessary, normal Tier Core control is suspended and the system transitions to:

Emergency Safety Mode

The Kill-Switch functions as the final protective safety layer within URSAS.

4.8 Relationship Between Warning Reason and Warning Intensity — Tier × Risk Structure

URSAS separately evaluates:

Tier = Warning Reason (Why to Warn)

and

Risk = Warning Intensity (How Strongly to Warn)

The Tier Engine determines the warning reason, while the Risk Evaluator determines the warning intensity.

The Tier Core then integrates both and determines the final warning policy.

Therefore, even within the same Tier, the Risk Level may differ depending on the evaluation performed by the Risk Evaluator.

This structure aims to avoid unnecessarily startling passengers while still providing sufficient warnings when needed.

This division of roles:

Why to Warn × How Strongly to Warn

forms the core of:

Tier-Based Safety Experience Control

Note: This relationship is organized in the Tier × Risk structure shown in **Fig. 4-3**.

4.9 Human Behavior Transition Model

The purpose of URSAS is not merely to issue warnings.

An important objective is to create preparation time for passengers to transition toward safer behavior and to support that behavioral change.

Through progressive warning levels:

Normal Operation
↓
Risk Level 1 — Attention
↓
Risk Level 2 — Caution
↓
Risk Level 3 — High Alert

URSAS supports a progressive behavioral transition from:

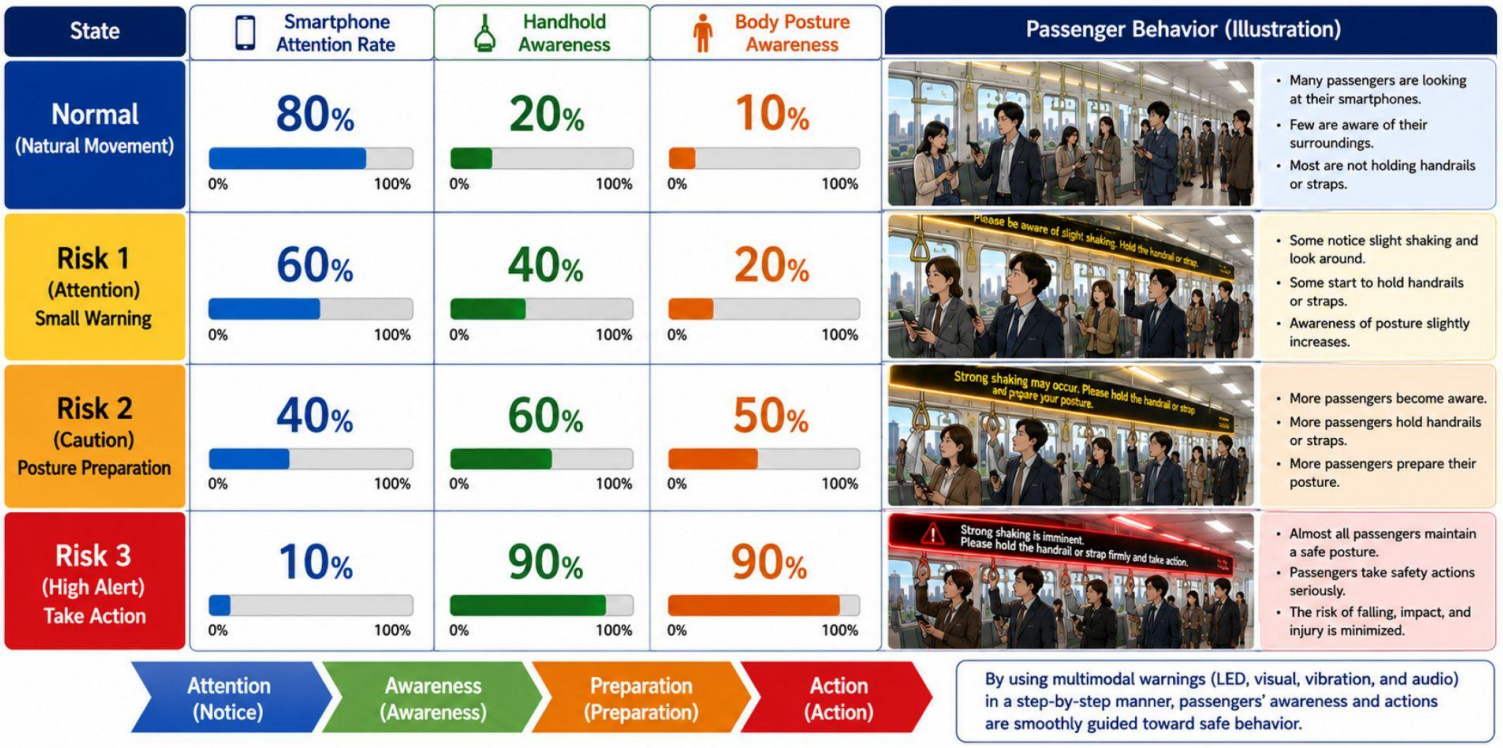
Focusing on a smartphone or other activity
↓
Awareness of surroundings
↓
Posture preparation
↓
Safety action

Through this progression, URSAS aims to reduce risks such as:

- falls;
- unintended contact with other passengers; and
- cascading loss of passenger stability inside the vehicle.

Figure 4-8: Passenger Behavior Transition Model (Human Behavior Transition Model)

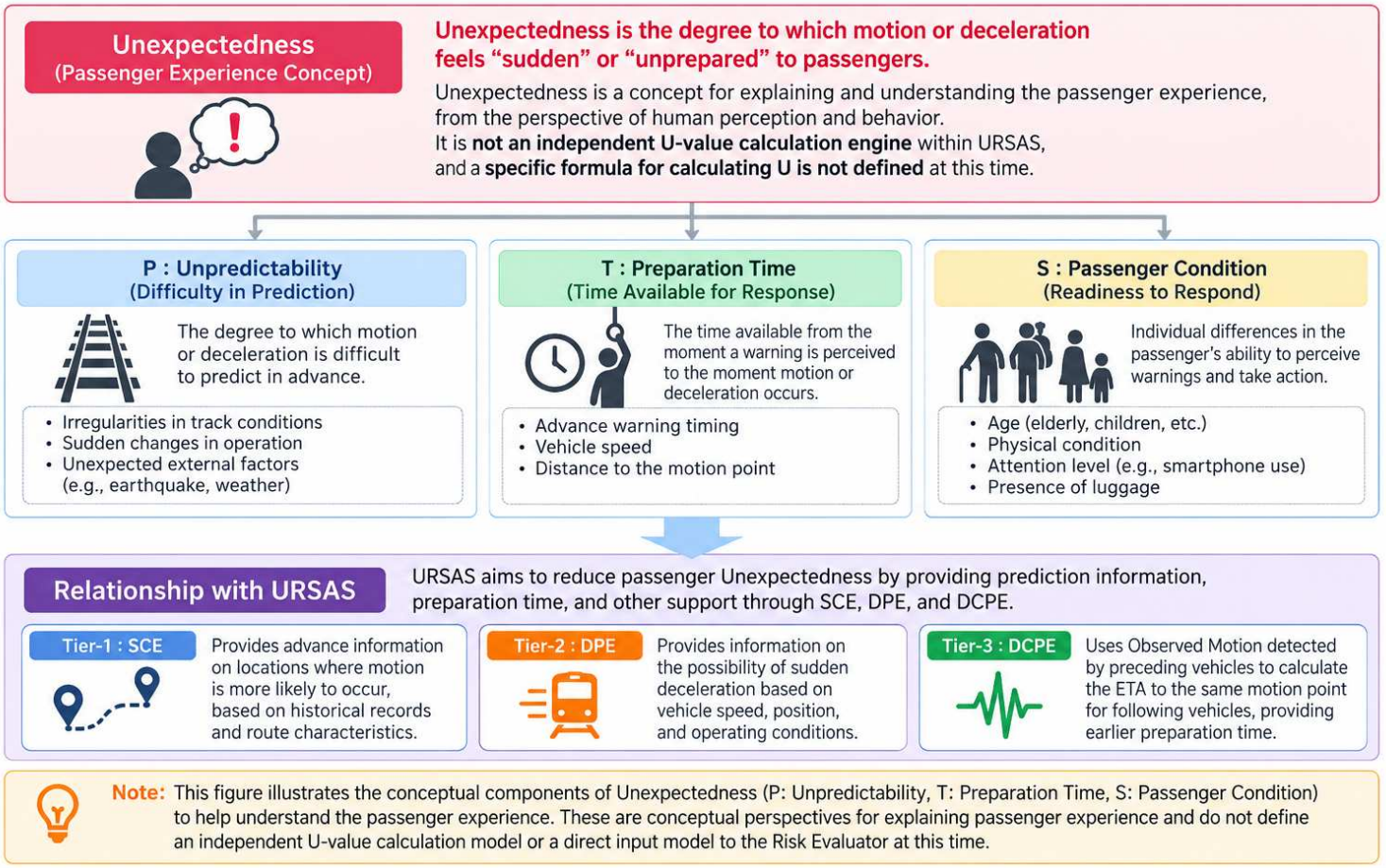
The figure below shows how passengers' awareness and actions change in response to step-by-step warnings.



Note: This figure presents a conceptual model of passenger behavior. The percentages shown are illustrative examples only.

Fig. 4-9 : Unexpectedness (Passenger Experience Concept)

Concept of Unexpectedness and Its Relationship with URSAS



URSAS is therefore designed not as:

“a system that simply warns passengers of danger,”

but as:

“a system that provides passengers with the time and information needed to take safety action.”

Chapter 5: Route Motion-Risk Information-Sharing Engine

SCE (Statistical Complement Engine)

5.1 What Is SCE?

SCE (Statistical Complement Engine) is the statistical risk-sharing engine responsible for Tier 1 within URSAS.

The defining feature of SCE is that all URSAS-equipped vehicles share and accumulate in the cloud the locations where motion was observed and the intensity of that motion during daily operation, and SCE uses this statistical information to generate risk-related information for Tier 1.

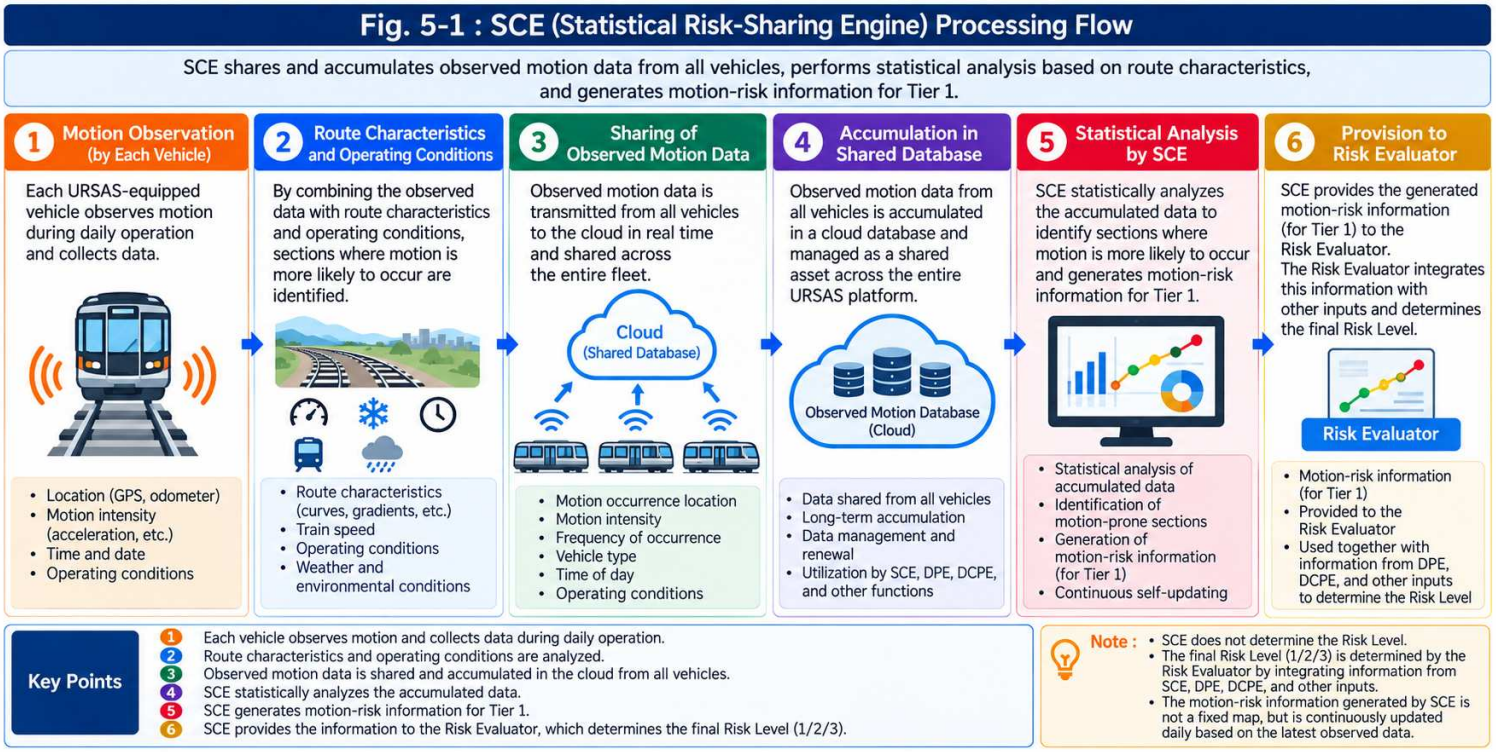
In other words, SCE is not a system that determines risk based on a single vehicle alone.

Motion information collected by all vehicles during daily operation is aggregated into a shared database, allowing all vehicles to share information such as:

“Motion is more likely to occur at this location.”

As a result, motion experienced by one vehicle can contribute to improving the safety of vehicles operating over the same route on subsequent days.

SCE statistically analyzes this large volume of accumulated data and generates Tier-1 risk-related information used in advance-warning decisions to help secure sufficient preparation time for passengers.



5.2 Daily Updated Motion-Risk Sharing System

SCE does not rely on a fixed risk map created once and used indefinitely.

Instead, the risk map is continuously updated using observation data transmitted by all vehicles operating each day.

Information shared by each vehicle may include:

- location where motion occurred;
- motion intensity and related measurement data;
- frequency of occurrence;
- route-specific characteristics;
- vehicle type;
- time of day; and
- operating conditions.

In addition, warning histories generated through DPE and DCPE, together with actual motion information exceeding defined reference thresholds, are accumulated within SCE.

SCE therefore functions as:

the shared database of the entire URSAS platform.

This database is not static.

For example, if track settlement, rail displacement, or another infrastructure-related condition is corrected through maintenance or repair, vehicles operating through that section on subsequent days may record little or no significant motion.

As a result, the risk-related information generated by SCE is continuously reassessed on the basis of new observation data.

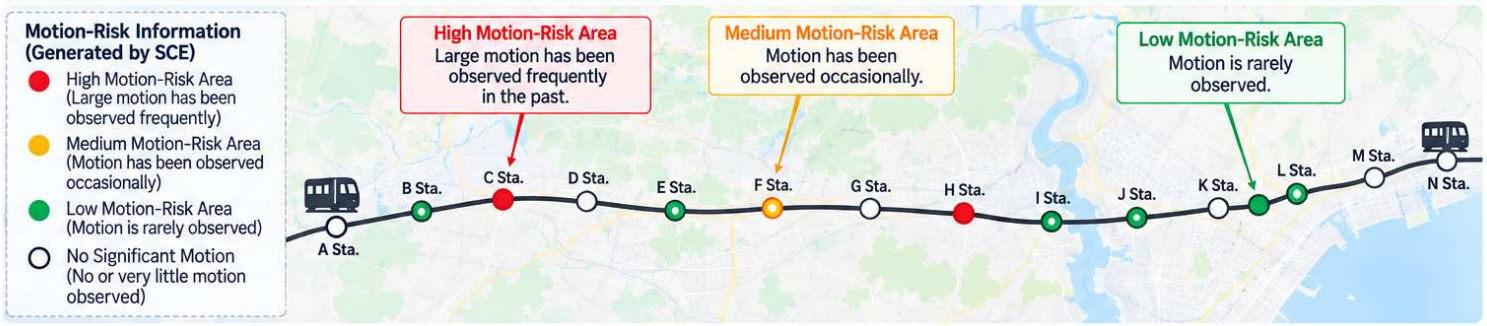
If significant motion is no longer observed over a defined period, the associated warning will gradually weaken and eventually disappear.

Conversely, if significant motion continues to be observed at a particular location, the risk-related value associated with that location will increase accordingly.

In this way, SCE functions as a continuously self-updating risk-sharing system that shares the latest field conditions across all vehicles.

Fig. 5-2 : Example of Route Motion-Risk Information Map (Generated by SCE)

This map is an example of motion-occurrence tendencies analyzed statistically by SCE.
It is continuously updated using daily observation data from all vehicles to provide motion-risk information for Tier 1.



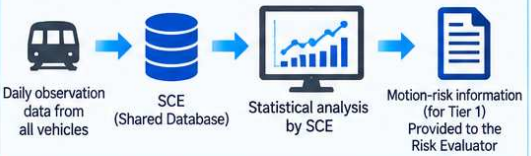
Applications (Examples)

- Support advance warnings and passenger preparation
- Improve passenger awareness and safety behavior
- Support efficient maintenance and targeted improvements
- Contribute to stable train operations and operational reliability

Overview of Motion-Risk Information (Example)

Section	Motion-Risk Level (SCE)	Main Factors (Examples)	Remarks
A – B	Low	Straight section, stable track conditions	Little motion observed
B – C	High	Curve, gradient, past frequent motion	Large motion observed frequently
C – D	Medium	Curve, weather influence	Occasional motion
D – E	Low	Straight section, stable track conditions	Little motion observed
E – F	Medium	Station approach, operating conditions	Occasional motion
F – G	High	Curve, track structure characteristics	Large motion observed frequently
…	…	…	…

Flow of Utilizing SCE Motion-Risk Information



Main Data Used for Analysis (Examples)

- Motion occurrence location
- Motion intensity
- Frequency of occurrence
- Route characteristics
- Vehicle type
- Time of day
- Operating conditions
- Warning histories (from DPE/DCPE)
- Actual motion data (exceeding reference thresholds)

Note :

- This map shows examples of motion-occurrence tendencies based on SCE statistical information.
- The actual warning intensity (Risk Levels 1–3) is determined by the Risk Evaluator by integrating information from SCE, DPE, and DCPE together with the current operating status and other operational factors.

5.3 Why SCE Supports the Entire URSAS Architecture

DPE and DCPE provide more specific prediction and analysis information, but each has limitations in its role.

For example, DCPE can provide the information required for advance-warning decisions for following vehicles only after Observed Motion has actually been detected by a preceding vehicle.

Therefore, DCPE cannot be used for:

- the leading vehicle; or
- the vehicle that first encounters the motion.

In addition, depending on vehicle speed and the distance between vehicles, sufficient preparation time may not always be available.

SCE compensates for these limitations.

By using the large volume of historical motion data accumulated over time, SCE can provide risk-related information before the motion is actually encountered.

This information is used in evaluation by the Risk Evaluator and supports warnings corresponding to Risk Levels 1 through 3.

SCE therefore functions as:

“the foundational system that provides early risk-related information to all vehicles.”

5.4 Characteristics and Limitations of SCE

The information used by SCE is based on statistically accumulated motion histories actually observed by individual vehicles.

For this reason, even at the same location, the characteristics and timing of motion may vary depending on factors such as:

- vehicle type;
- train formation length;
- passenger load;
- vehicle weight;
- weather conditions; and
- maintenance conditions.

In addition, the timing of sudden deceleration may vary depending on current operating conditions, including service-delay recovery operations.

Therefore, SCE can determine that:

“Motion is more likely to occur at this location,”

but it cannot accurately predict:

“Exactly how many seconds remain before the motion occurs.”

For this reason, SCE provides early risk-related information that supports advance warnings before the expected occurrence of motion.

This is a deliberate design principle intended to provide passengers with sufficient preparation time, and SCE therefore functions as a Tier-1 engine focused on early precautionary information.

Cases SCE Cannot Address Independently

Because SCE is based on a statistical model, it also has inherent limitations.

For example, SCE cannot generate advance risk-related information based on prior observations for events that have never previously been observed, such as:

- track settlement that has occurred for the first time that day;
- sudden rail displacement; or
- rapid changes in track conditions caused by natural disasters or other unexpected events.

SCE also has difficulty providing highly accurate timing predictions for train-specific operating conditions, such as:

- the exact onset of sudden deceleration;
- service-delay recovery operations; and
- operating conditions specific to an individual train formation.

These limitations cannot be fully addressed by SCE alone and are therefore complemented through integration with DPE and DCPE.

5.5 Division of Roles Among SCE, DPE, and DCPE

The roles of the three engines are divided as follows:

- **SCE** shares information about locations where vehicle motion is more likely to occur, based on accumulated historical motion data and route characteristics.
- **DPE** predicts the possibility of sudden deceleration in the near future, based on current operating conditions and historical data.
- **DCPE** calculates ETA based on Observed Motion actually detected by a preceding vehicle and provides advance information for the Following Vehicle immediately behind it.

SCE provides broad and continuous advance information based on accumulated historical data.

DPE complements SCE by predicting sudden deceleration based on current operating conditions.

DCPE further complements both systems through precisely timed information based on actual Observed Motion and ETA calculations.

By combining these three approaches, the strengths of each engine compensate for the limitations of the others, contributing to a highly reliable passenger warning system.

Within DCPE, each Warning-Responsible Vehicle is responsible only for the single Following Vehicle immediately behind it.

This one-vehicle responsibility structure allows the current operating conditions at each stage of the train formation to be reflected in ETA calculations while helping maintain warning accuracy as information is passed rearward.

Chapter 6: Deceleration Prediction Engine

DPE (Deceleration Prediction Engine)

6.1 Overview

The Deceleration Prediction Engine (DPE) is the Tier 2 prediction engine within URSAS.

DPE is designed primarily to reduce passenger stress in public transportation, particularly the anxiety associated with unexpected deceleration in crowded vehicles, and to help create an environment in which passengers can travel with greater confidence and comfort.

URSAS is not a system for controlling railway operations.

Nor is it a system intended to change how train drivers operate their vehicles.

The objective of URSAS is:

“to enhance safety and comfort by providing each passenger with relevant information slightly in advance.”

In public transportation, many passengers travel while standing, particularly during periods of high demand.

In such environments, the greatest source of anxiety, tension, or loss of balance may not be deceleration itself, but the fact that:

deceleration begins unexpectedly.

To reduce this element of unexpectedness, DPE analyzes current train-operating conditions and shares predictive information such as:

“Deceleration may occur soon.”

The objective is not to predict the exact moment of deceleration with absolute precision.

Rather, it is to provide passengers with a few seconds of preparation time, allowing them to prepare naturally for the next movement.

DPE therefore serves Tier 2 within URSAS not only as:

“a deceleration prediction engine,”

but also as:

“an engine that provides reassurance through preparation time.”

6.2 The Problem DPE Seeks to Address

In public transportation, deceleration occurs in many situations, including station approaches, signal-related operations, and speed-restricted sections.

Under normal operating conditions, deceleration is generally performed in a relatively gradual manner, and significant passenger instability may be limited.

However, when a train experiences a minor delay due to factors such as increased passenger boarding time or delayed door closure, operational adjustments may sometimes result in portions of the journey being operated differently in an effort to recover service time.

In such circumstances, the remaining distance available for deceleration may become shorter, causing braking to begin closer to the stopping point.

This can require greater deceleration over a shorter distance and may increase the inertial forces experienced by standing passengers.

In crowded vehicles, these forces can contribute to a cascading loss of balance, in which passengers are displaced in the same direction and instability spreads through the surrounding passenger area.

This phenomenon may increase not only the risk of falls and unintended contact between passengers, but also feelings of crowding, pressure, and psychological stress.

DPE is not intended to eliminate such events completely.

As a prediction engine, it cannot guarantee that its warning will exactly match the actual start of deceleration.

For this reason, warnings are intentionally issued somewhat early, with priority given to passenger safety.

Even this brief preparation time can allow passengers to naturally grasp a handrail, strap, or other support and stabilize their posture.

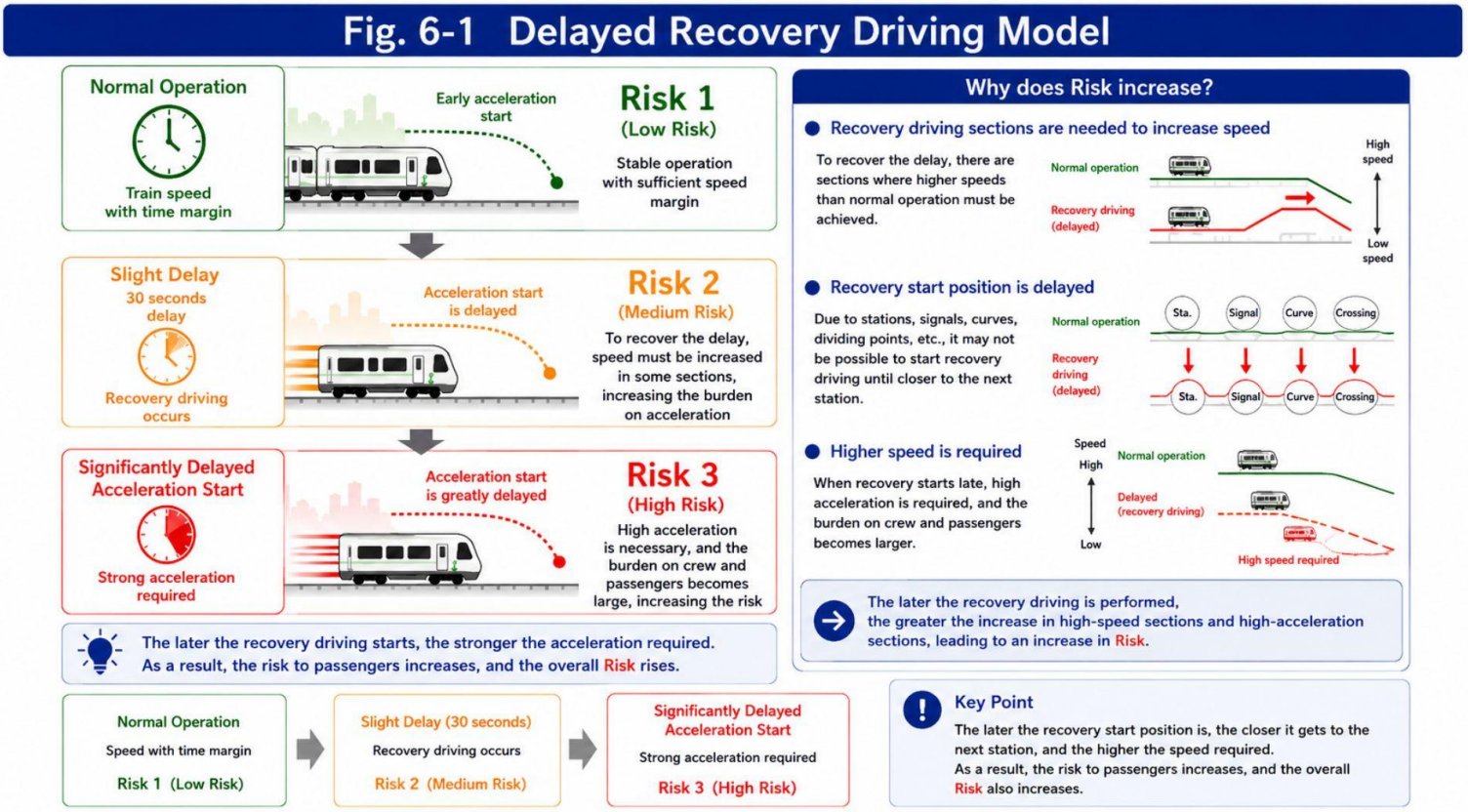
If even some passengers begin supporting themselves in advance, it may also help limit the spread of cascading instability within a crowded vehicle.

The problem DPE seeks to address is therefore not simply:

“predicting deceleration,”

but rather:

“reducing the anxiety, stress, and fall risk caused by unexpected deceleration, and creating an environment in which passengers can use public transportation with greater confidence and comfort.”



6.3 Role as Tier 2

To support this passenger experience, DPE is positioned within URSAS as Tier 2 — near-future prediction.

URSAS uses a three-layer prediction architecture:

- Tier 1 — SCE
- Tier 2 — DPE
- Tier 3 — DCPE

Within this architecture, DPE predicts the possibility of upcoming deceleration based on information such as train speed and operating conditions.

When DPE detects conditions associated with likely deceleration, it generates predictive deceleration information and provides this information to the Risk Evaluator.

While Tier 1 provides risk-related information based on historical motion data, Tier 2 provides a more immediate predictive indication:

“Deceleration may occur soon.”

DPE is therefore positioned as:

a prediction engine that estimates the likelihood of upcoming deceleration and provides predictive deceleration information to the URSAS decision architecture.

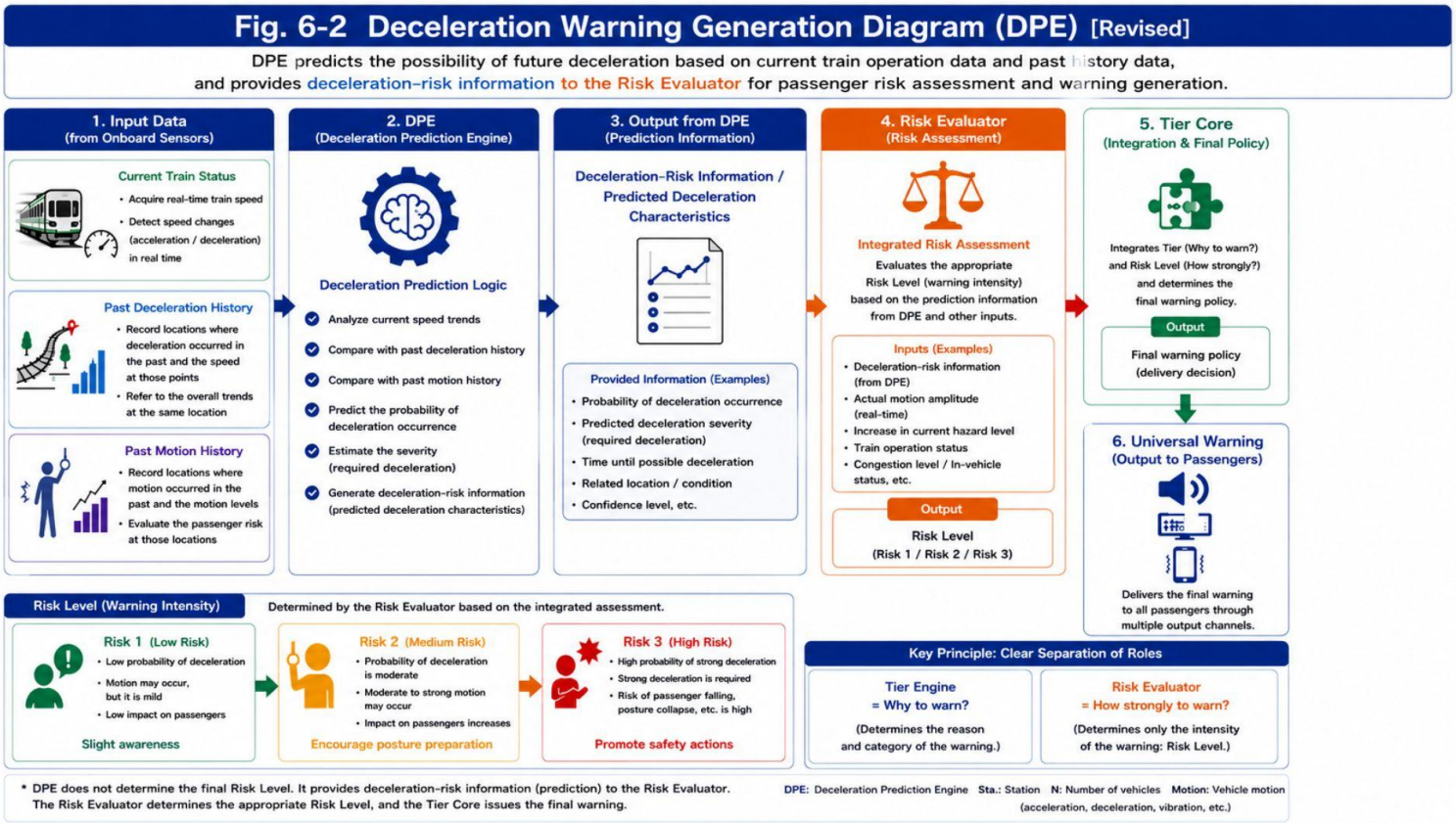
6.4 Input Information

DPE integrates multiple sources of operational data to evaluate the possibility of deceleration occurring in the near future.

Primary input information includes:

- train speed;
- acceleration;
- GPS position data;
- ATS position information;
- stopping-position information;
- historical deceleration data;
- historical motion data; and
- operating conditions, including delay status.

By analyzing these inputs together, DPE estimates the likelihood and characteristics of upcoming deceleration and provides this information to the Risk Evaluator for passenger-risk assessment.



6.5 Outputs

DPE prediction outputs are classified as Tier 2 information and are provided to the URSAS decision architecture for warning control.

Primary output methods include:

- large displays and flashing yellow LED lines;
- handrail / support-bar vibrators;
- floor vibration devices;
- a brief audible notification;
- smartphone notifications via BLE.

These outputs are intended to:

encourage passengers to prepare their posture, provide sufficient time to prepare for deceleration, and reduce anxiety and stress in crowded public transportation environments.

6.6 The Essence of DPE

DPE is not merely a technology for predicting deceleration.

Its essential purpose is:

“to provide passengers with reassurance through preparation time.”

Much of the stress experienced in crowded public transportation does not necessarily arise from the magnitude of motion itself, but from the fact that it occurs unexpectedly.

Even a warning only a few seconds in advance can allow passengers to naturally adjust their posture and hold onto a handrail, strap, or other support.

This brief preparation time can do more than reduce the risk of falling.

It can also reduce anxiety and tension and help passengers feel more secure.

In this sense, what DPE ultimately provides is not merely:

deceleration prediction,

but:

reassurance.

The accumulation of these moments of reassurance can help transform public transportation from:

“a space used only for movement”

into:

“a public transportation environment that people can use with confidence, safety, and comfort.”

This is the fundamental reason for DPE’s existence and one of the core values of the Safety Experience envisioned by URSAS.

Chapter 7: Deterministic Chain Prediction Engine

DCPE (Deterministic Chain Prediction Engine)

7.1 Overview

URSAS aims to use the ETA calculated by DCPE to provide advance information approximately two to three seconds before a vehicle reaches the Motion Point, giving passengers time to stabilize their posture.

To secure this preparation time of approximately two to three seconds, DCPE uses the V-Referencer to select an appropriate Reference Vehicle according to vehicle speed and other operating conditions.

For example, when vehicle speed is high, selecting the immediately preceding vehicle as the Reference Vehicle may leave insufficient time before the Following Vehicle reaches the Motion Point.

For this reason, the V-Referencer selects an appropriate vehicle ahead—such as the vehicle one car ahead, two cars ahead, or, in some cases, three cars ahead—as the Reference Vehicle in order to secure approximately two to three seconds of preparation time.

On the other hand, because vehicles accelerate and decelerate while in operation, directly calculating an ETA far in advance for a distant vehicle may result in increasing ETA error as vehicle speed changes.

For example, if Observed Motion detected by the leading vehicle is used to directly predict that a vehicle 12 cars behind will reach the Motion Point approximately 10.8 seconds later, subsequent deceleration—such as when approaching a station—may cause the actual arrival time to be later than the initially calculated ETA.

To reduce the accumulation of ETA error associated with such long-range predictions, DCPE adopts a Chain Responsibility Structure in which each Warning-Responsible Vehicle is responsible only for the single Following Vehicle immediately behind it.

Each Warning-Responsible Vehicle calculates and updates the ETA for the Following Vehicle immediately behind it using the latest available vehicle-speed information, together with the Observed Motion and Motion Point information shared from the Reference Vehicle.

In this way, DCPE uses the V-Referencer to select a Reference Vehicle that helps secure approximately two to three seconds of preparation time, while the Chain Responsibility Structure helps reduce the accumulation of ETA error caused by changes in operating conditions such as acceleration and deceleration.

7.2 Purpose of DCPE

The purpose of DCPE is not to estimate motion that may occur.

Its purpose is:

“to share motion that has actually occurred.”

SCE uses statistical information.

DPE predicts the possibility of deceleration.

DCPE, by contrast, uses:

actual observed motion itself.

For this reason, DCPE provides the highest level of confidence among the three URSAS prediction layers.

7.3 Role as Tier 3

URSAS uses a three-layer prediction architecture:

- Tier 1 — SCE
- Tier 2 — DPE
- Tier 3 — DCPE

DCPE is responsible for:

Tier 3

Tier 3 is the layer that provides advance warning for motion that has already been confirmed by actual observation.

Because significant motion has already been observed, Tier 3 is intended to encourage passengers to take clear safety action.

DCPE is therefore positioned as:

the engine that supports the final stage of passenger safety action.

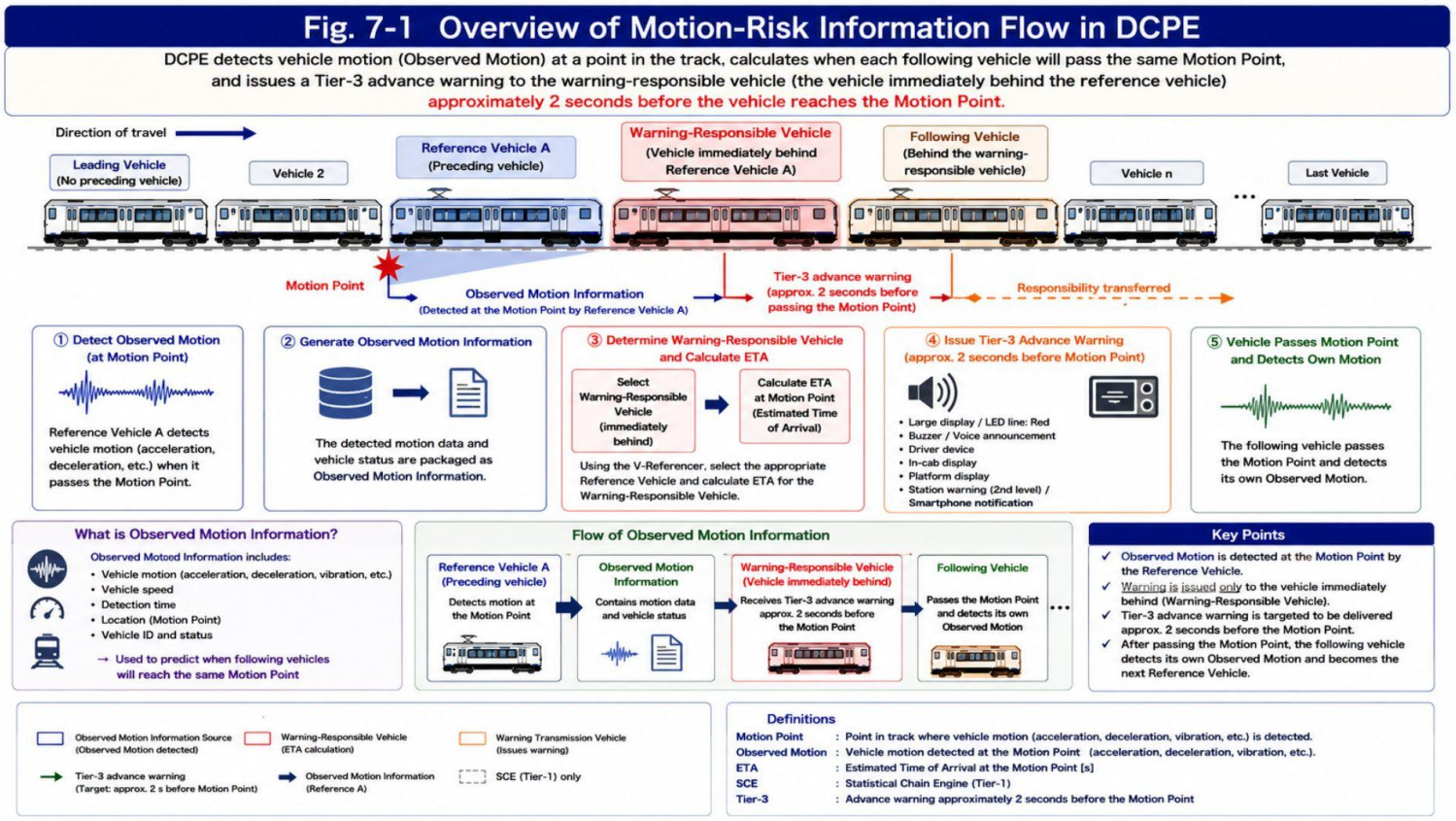
7.4 Input Information

DCPE performs its evaluation using real-time information.

Primary input data includes:

- IMU data;
- acceleration data;
- angular velocity data;
- direction-of-travel information;
- vehicle position information;
- train speed information; and
- motion information received from preceding vehicles.

These inputs are integrated to evaluate the timing and location of the observed motion event relative to following vehicles.



7.5 Reference-Vehicle Selection Using the V-Referencer

DCPE uses the:

V-Referencer

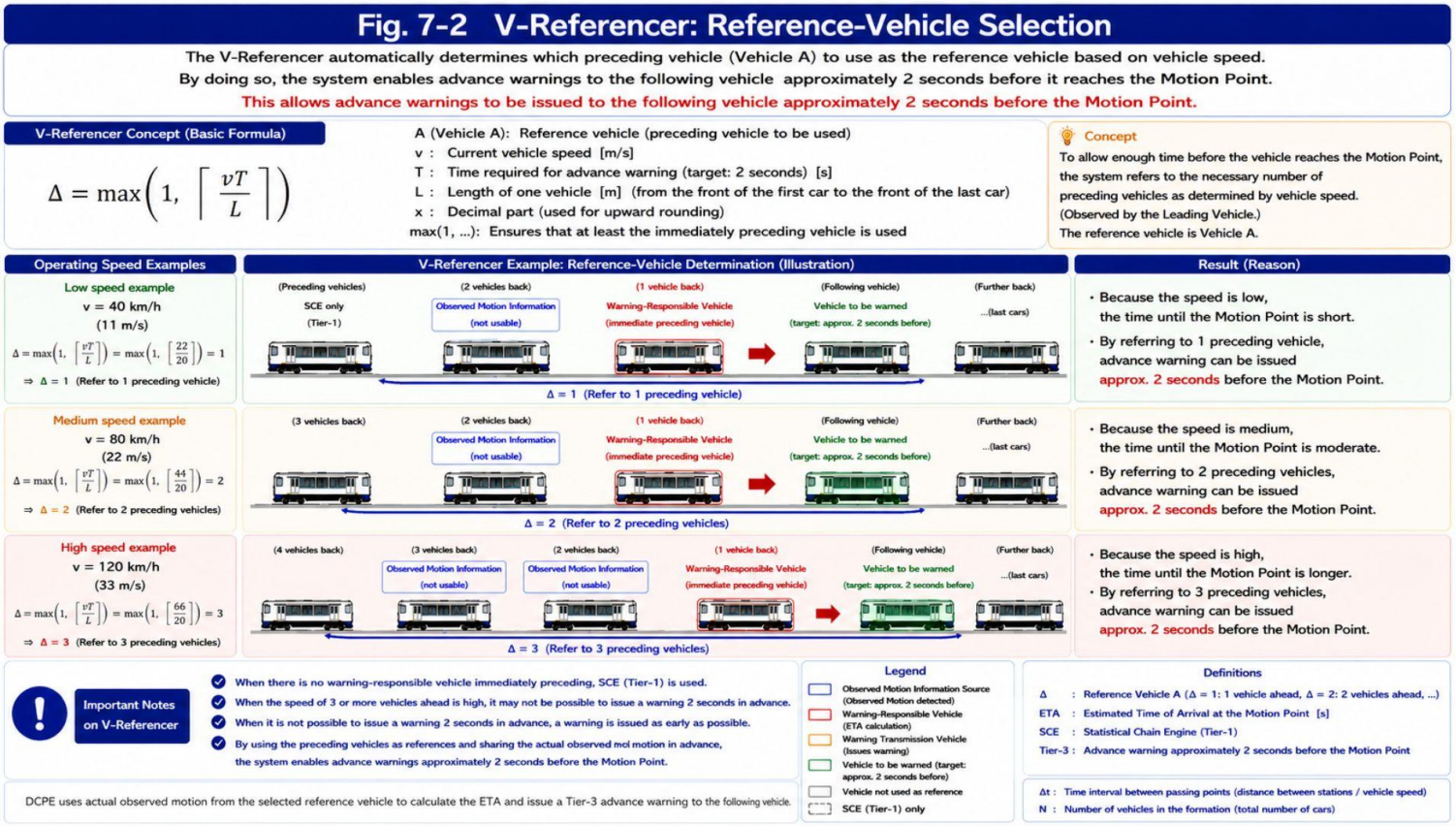
to determine which preceding vehicle should be used as the reference vehicle.

The reference vehicle is dynamically changed according to train speed.

At lower speeds, DCPE refers to a nearby preceding vehicle.

At higher speeds, it may refer to a vehicle farther ahead in order to optimize available warning time.

This dynamic selection enables flexible warning control that responds to changes in train speed.



7.6 ETA Calculation

DCPE calculates the:

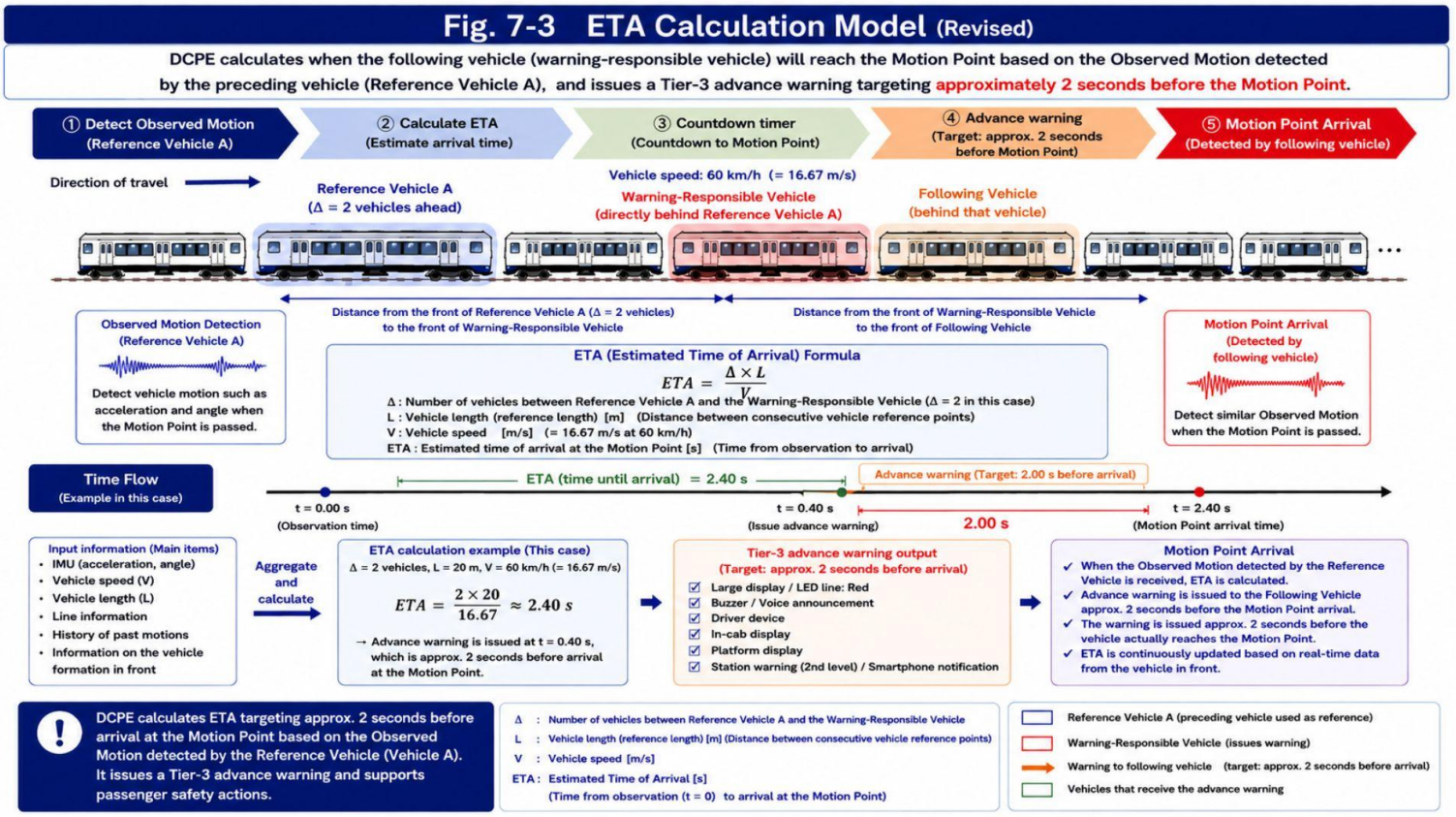
Estimated Time of Arrival (ETA)

at the motion point for a following vehicle, using motion information observed by the selected reference vehicle.

The ETA represents:

“approximately how many seconds remain before the following vehicle reaches the motion point.”

URSAS uses this ETA to determine the timing of advance warnings.



7.7 Outputs

DCPE outputs are classified as Tier 3 information and are provided to the URSAS decision architecture for final warning control. The resulting passenger notifications are delivered through Universal Warning using appropriate output devices.

Primary output methods include:

- large displays and flashing red LED lines;
- handrail / support-bar vibrators;
- floor vibration devices;
- projector warning displays;
- two brief audible notifications; and
- smartphone notifications.

These outputs are intended to encourage passengers to take clear and immediate safety action.

7.8 Differences Between SCE, DPE, and DCPE

DCPE serves a different role from SCE and DPE.

SCE shares potential risk based on historical statistical information.

DPE predicts near-future deceleration based on current operating conditions.

DCPE provides real-time observed-motion and ETA information based on motion that has already been observed.

In simple terms:

SCE = Past

DPE = Near Future

DCPE = Real Time

URSAS adopts a hierarchical chain-of-responsibility structure in which each vehicle is responsible only for warning the single vehicle immediately behind it.

This makes it possible to transmit warning information rearward while maintaining a simple and highly reliable system structure, even in long train formations.

It also eliminates the need for a single leading vehicle to communicate directly with every vehicle in the train, helping reduce system load and communication latency.

In addition, by assigning each Warning-Responsible Vehicle responsibility for only the vehicle immediately behind it, URSAS seeks to minimize errors in ETA calculation.

For example, it is technically possible to use motion data observed by the leading vehicle to directly estimate when a vehicle ten cars behind will reach the same location.

However, if the train accelerates or decelerates during the interval before that vehicle reaches the motion point, prediction error may increase as the distance and number of vehicles involved become greater.

This effect may be particularly significant in long train formations, where changes in train speed and operating conditions can accumulate over time and reduce the accuracy of ETA estimates for distant vehicles.

For this reason, URSAS uses a method in which each vehicle is responsible only for the vehicle immediately behind it and recalculates ETA using the most recent speed information and observed motion data available at that point.

Through this chained reassessment process, the influence of speed changes can be localized, helping maintain higher prediction accuracy even in long train formations.

The chain-of-responsibility structure therefore serves not only to improve communication efficiency, but also to help preserve ETA accuracy.

7.9 Chain Responsibility Structure

DCPE adopts a Chain Responsibility Structure in which each Warning-Responsible Vehicle is responsible for issuing an advance warning only to the vehicle immediately behind it.

Observed Motion detected by a preceding vehicle is not simply broadcast uniformly to all vehicles in the formation. Instead, DCPE uses the V-Referencer to select an appropriate Reference Vehicle based on the current operating conditions and the time required for advance warning.

Using the Observed Motion detected by the selected Reference Vehicle, DCPE calculates the ETA (Estimated Time of Arrival at the Motion Point) for the target vehicle. This information is then used to support an advance warning before the vehicle reaches the same Motion Point.

The responsibility for warning is transferred sequentially through the formation:

Reference Vehicle → Warning-Responsible Vehicle → Following Vehicle

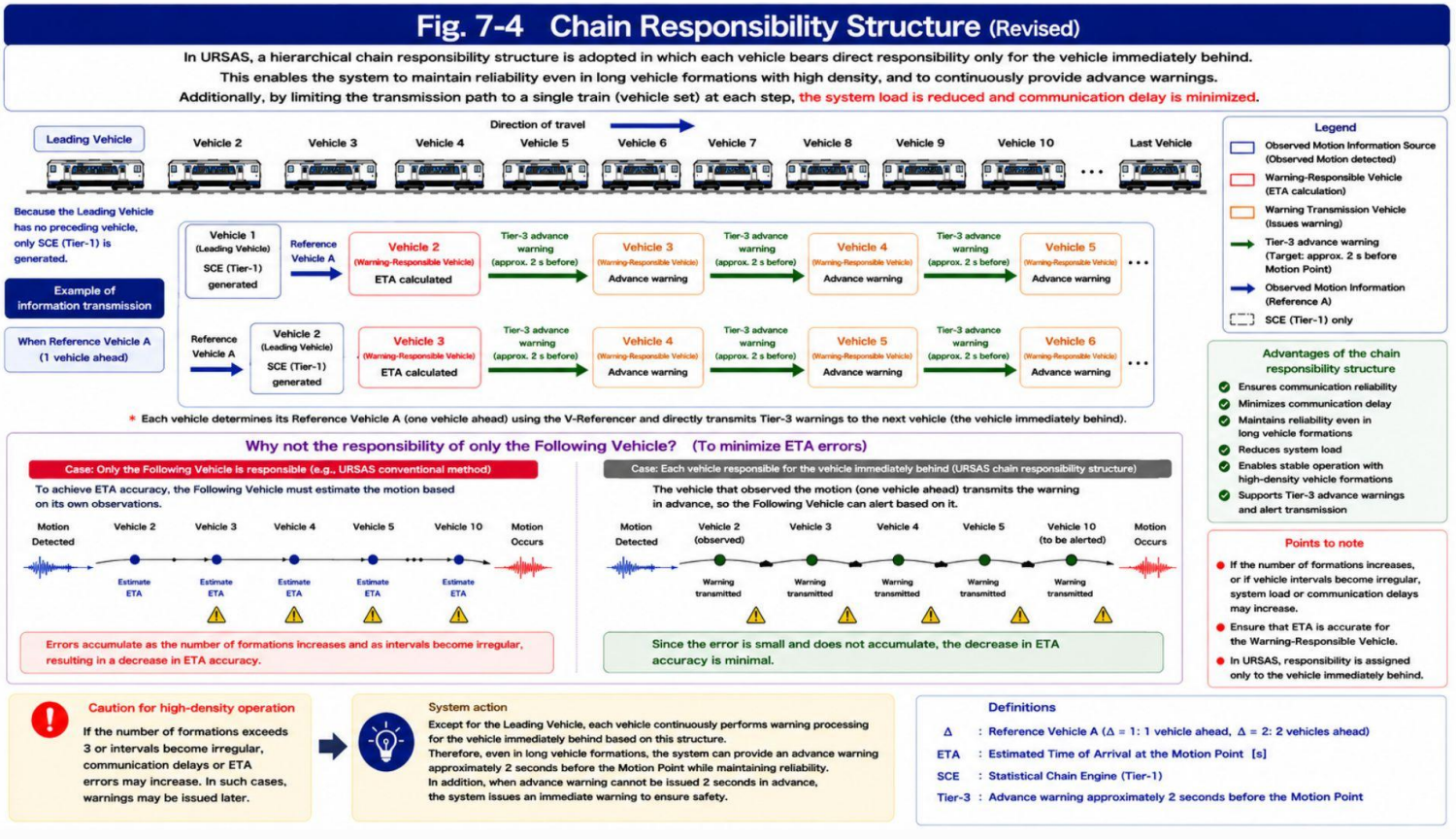
After the Following Vehicle passes the Motion Point and obtains its own Observed Motion data, it can become the next Warning-Responsible Vehicle for the vehicle immediately behind it.

In this way, warning responsibility is passed from one vehicle to the next rather than being assigned simultaneously to the entire formation.

This structure has three main purposes:

- to clearly define which vehicle is responsible for warning which following vehicle;
- to prevent unnecessary or duplicated warnings from being distributed throughout the formation; and
- to allow warning information to be continuously updated using the latest Observed Motion data as vehicles pass the Motion Point.

As a result, DCPE forms a continuous responsibility chain in which each vehicle contributes to the safety of the vehicle immediately behind it.



7.10 Significance of DCPE

DCPE is the final prediction layer within URSAS.

SCE supports early awareness.

DPE encourages posture preparation.

DCPE supports clear safety action.

What DCPE handles is not merely a possibility.

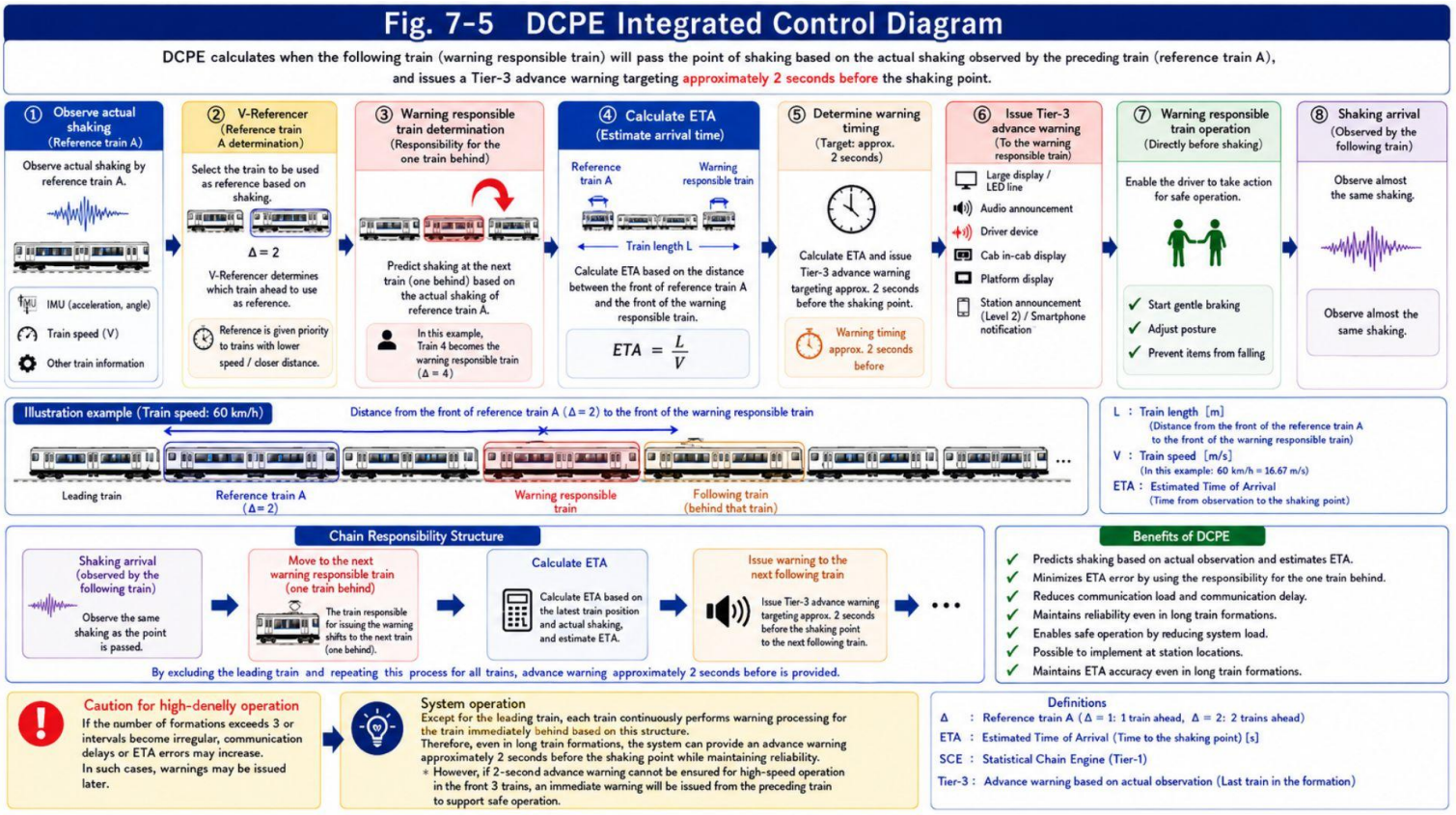
It handles:

motion that has actually been observed.

For this reason, DCPE functions as the highest-confidence warning engine within URSAS.

By providing passengers with a few seconds of preparation time, DCPE aims to reduce risks such as:

- falls;
- unintended contact between passengers; and
- dropped personal belongings.



Chapter 8: Safety Display Platform and Advertising Utilization

In URSAS, multiple output interfaces are combined to deliver information reliably to passengers.
This section explains the main daily passenger notification and warning interfaces.



The upper monitor area (approx. 40 cm) displays information such as shaking warnings, operation information, advertisements, and G-Sync status.
The LED line below (approx. 10 cm) changes color and blinking patterns according to the risk level to encourage passengers to pay visual attention.

8.1 Overview

The large onboard display used in URSAS is not primarily an advertising medium.

Its fundamental purpose is to serve as a:

Safety Display Platform

for reliably communicating safety information to passengers.

The large display is designed as an important URSAS safety device responsible for functions such as:

- safety warning displays;
- LED line indications;
- evacuation guidance;
- general safety information.

URSAS also makes effective use of this safety infrastructure during normal operation to provide services such as advertising, tourism information, and multilingual passenger information.

8.2 Safety Display Platform — Large Onboard Display

The large onboard display is one of the core devices that make up Universal Warning.

During normal operation, it provides onboard information and passenger guidance.

When Tier 1, Tier 2, or Tier 3 is activated, the display immediately switches to the appropriate safety warning mode.

The display also operates in coordination with:

- LED lines;
- projectors;
- audible notifications; and
- vibration-based notifications.

By combining these channels, URSAS provides multimodal warnings through:

- visual;
- auditory; and
- tactile

communication.

The objective is to create an environment in which essential safety information can be understood intuitively by passengers regardless of nationality, age, or disability.

8.3 Advertising Using Safety Infrastructure

In URSAS, large displays installed primarily for safety purposes can also be used as advertising media during normal operation.

Traditional suspended and printed onboard advertising requires recurring operational processes such as:

- printing;
- transportation;
- replacement; and
- disposal.

By transitioning to large digital displays, operators can support:

- video advertising;
- multilingual advertising;
- time-of-day-based advertising;
- location-linked advertising; and
- tourism information.

Advertising content can also be updated remotely, significantly improving operational flexibility and efficiency.

8.4 Evolution Beyond Traditional Onboard Advertising

The introduction of large digital displays makes it possible to substantially reduce or potentially replace conventional printed onboard advertising.

Expected benefits include:

- reduced manual advertising replacement work;
- lower printing costs;
- reduced material waste; and
- reduced workload for operational staff.

In addition, the entire vehicle interior can be used as a coordinated visual environment, enabling forms of communication and presentation that are difficult to achieve with conventional static advertising.

8.5 Creating New Advertising Value

Large onboard displays make it possible to move beyond the traditional concept of fixed advertising spaces.

Potential examples include:

- trains themed around film releases;
- tourism promotion trains;
- corporate brand trains; and
- special event or charter advertising trains.

These formats can create high-value advertising opportunities that are difficult to achieve through conventional printed onboard media.

However, URSAS assumes that advertising must not compromise the quietness, comfort, or usability of the passenger environment.

Advertising should not interfere with the transportation experience.

Instead, it should be presented in a manner that harmonizes with the public transportation environment.

8.6 Contribution to Implementation and Operating Costs

Traditionally, safety infrastructure is often regarded primarily as a cost.

URSAS introduces a different approach.

By using large displays installed for safety purposes as advertising media during normal operation, part of the implementation and operating costs may be offset through advertising revenue.

This does not mean installing safety equipment for the purpose of advertising.

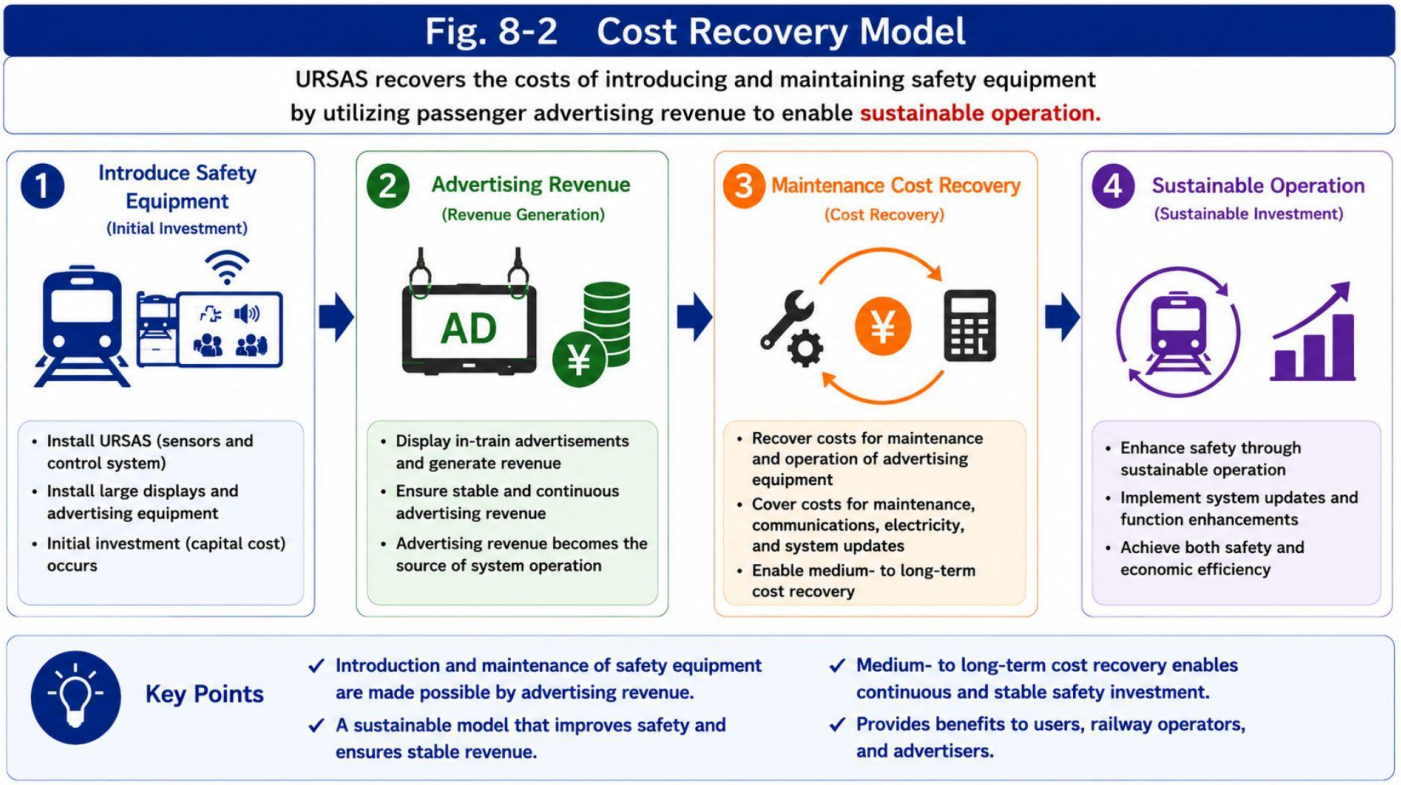
Rather, it means that:

infrastructure required for safety can also contribute to generating revenue that helps support its own implementation and maintenance.

Through this approach, URSAS seeks to achieve improvements in:

- safety;
- business sustainability; and

- long-term operational sustainability.



8.7 A Foundation for the Future of Public Transportation

The large onboard display is not only a safety information device.

It also serves as a foundation for future public transportation experiences.

The following functions, described in the next chapter, are built around this display platform:

- G-Sync, the immersive experience system;
- station geofencing; and
- interactive advertising.

URSAS begins with the goal of improving passenger safety, but ultimately seeks to contribute to the broader evolution of public transportation itself.

Chapter 9: Experiential Advertising and the Future of Public

Transportation

9.1 Overview

G-Sync (Geo-Synchronized Experience) is an immersive experience system that synchronizes large onboard displays, LED lines, audio, and other effects with stable and predictable acceleration and deceleration within designated station geofences.

This chapter explains how G-Sync can enhance the public transportation experience while remaining consistent with the safety-first philosophy of URSAS, helping transform public transportation from a “space for movement” into a “space for experience.”

URSAS seeks to create new value in public transportation through advertising and information presented on large onboard displays.

However, the objective of URSAS goes beyond conventional digital advertising.

This chapter describes the progression from:

Advertising

↓

Experiential Advertising

↓

G-Sync (Geo-Synchronized Experience)

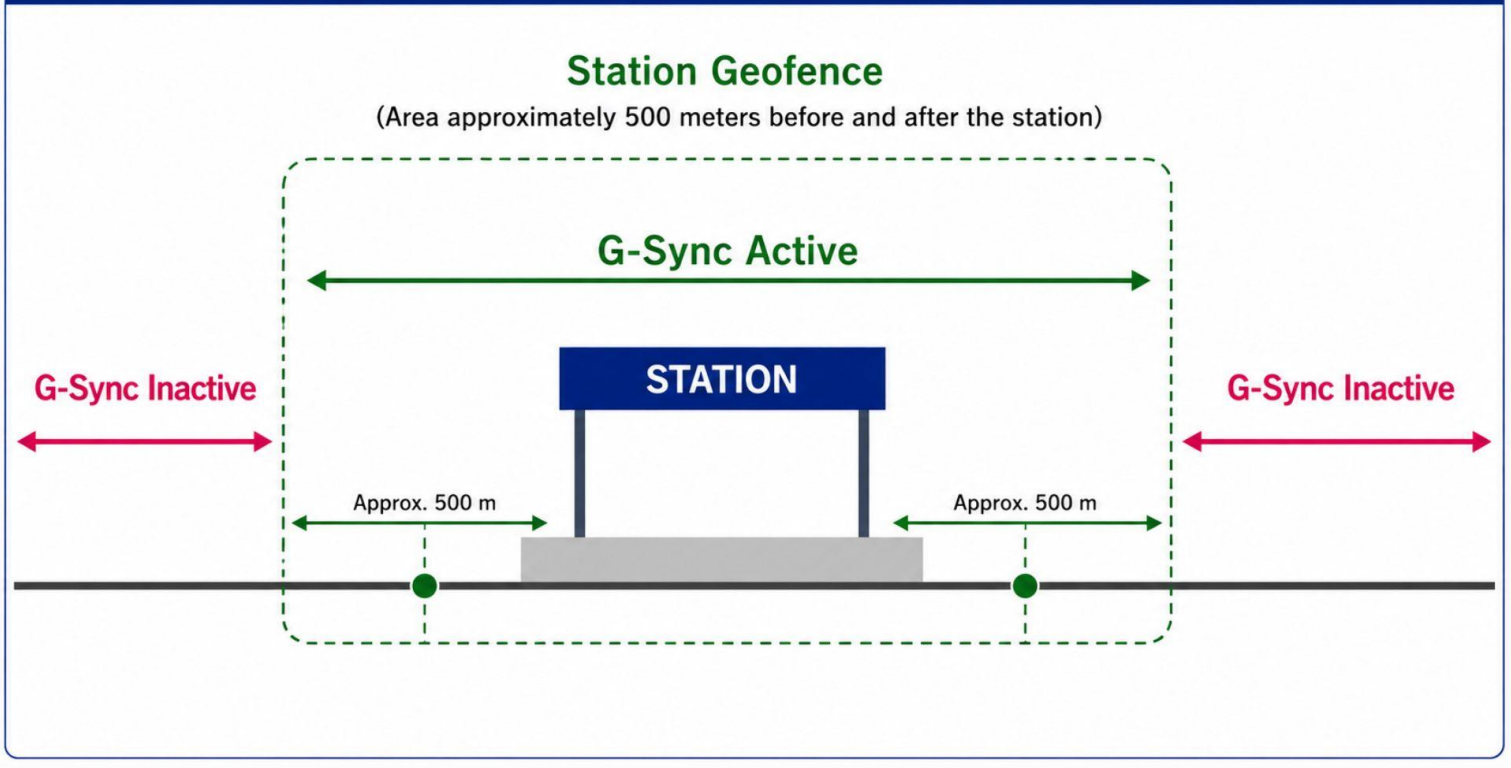
↓

Station Geofencing

↓

The Future of Public Transportation

Fig. 9-1 G-Sync Operating Range



9.2 From Advertising to Experiential Advertising

Traditional railway advertising has primarily been:

“advertising that passengers view.”

Passengers receive information by viewing posters, suspended advertisements, and other static media.

With the large onboard displays used by URSAS, however, new forms of advertising and passenger experiences can be created by combining:

- visual content;
- audio;
- LED lines; and
- vibration-based effects.

This allows advertising to evolve from:

“advertising that passengers view”

to:

“advertising that passengers experience.”

For example, the entire vehicle environment could be used to create experiences associated with:

- tourism promotion;
- film-release campaigns;
- regional promotion;
- corporate brand experiences.

9.3 Value Created by Experiential Advertising

Experiential advertising can provide a level of immersion that is difficult to achieve through conventional advertising.

Rather than simply viewing an advertisement, passengers can experience content in which visual effects are synchronized with predictable vehicle movement.

Experiential content can also be used to provide:

- tourism information;
- local and regional information; and
- event information.

This can enhance the value of the travel experience itself.

URSAS aims to develop public transportation not merely as an advertising medium, but as:

an experiential space.

9.4 Why Station Geofencing Is Necessary

URSAS uses station geofencing to enable experiential content while maintaining its safety principles.

Trains continuously accelerate and decelerate during operation.

However, many of these changes in motion may be difficult for passengers to anticipate and can therefore involve a degree of unexpectedness.

A fundamental principle of URSAS is:

Reducing Unexpectedness

For this reason, unpredictable acceleration or deceleration during normal travel should not be intentionally incorporated into experiential effects.

In contrast, acceleration and deceleration associated with routine station arrival and departure are generally more predictable from the passenger’s perspective.

These operating phases can also provide relatively stable patterns of vehicle motion suitable for controlled experiential applications.

URSAS therefore defines designated stable acceleration and deceleration zones around stations as:

Station Geofences

and limits the use of G-Sync to these designated areas.

This makes it possible to provide new experience value while preserving the fundamental safety philosophy of URSAS.

9.5 What Is G-Sync?

G-Sync (Geo-Synchronized Experience) is an immersive experience system that uses stable and predictable acceleration and deceleration within station geofences to synchronize:

- large onboard displays;
- LED lines;
- audio;
- vibration-based effects; and
- other compatible experience functions.

For example, as a train approaches a station, visual effects can begin in coordination with predictable deceleration, while LED lines and audio respond in synchronization.

Similarly, when the train departs from a station, the experience can transition to a different sequence synchronized with acceleration.

Through this approach, conventional advertising can evolve from:

“advertising that passengers view”

to:

“advertising that passengers experience.”

G-Sync is designed to transform part of the time spent using public transportation into a new form of passenger experience value.

9.6 Coexistence with Safety

Within URSAS:

Safety always takes priority over experience value.

If any of the following is activated:

- Tier 1;
- Tier 2;
- Tier 3; or
- Kill-Switch,

G-Sync is immediately suspended.

The large onboard displays also switch immediately from advertising or experiential content to the appropriate safety information.

This architecture enables experiential functions to coexist with safety without compromising the priority of safety-related communication.

Within URSAS:

Safety is always the highest priority.

9.7 The Future of Public Transportation

URSAS is not intended to be simply a warning system.

Nor is it intended to be simply an advertising system.

Its broader objective is to create a new form of public transportation that integrates:

Safety

Information

Experience

By naturally incorporating:

- local culture;
- tourism information; and
- brand experiences

into everyday moments such as station arrival and departure, travel time itself can gain additional value.

With safety as its foundation, URSAS aims to transform public transportation from:

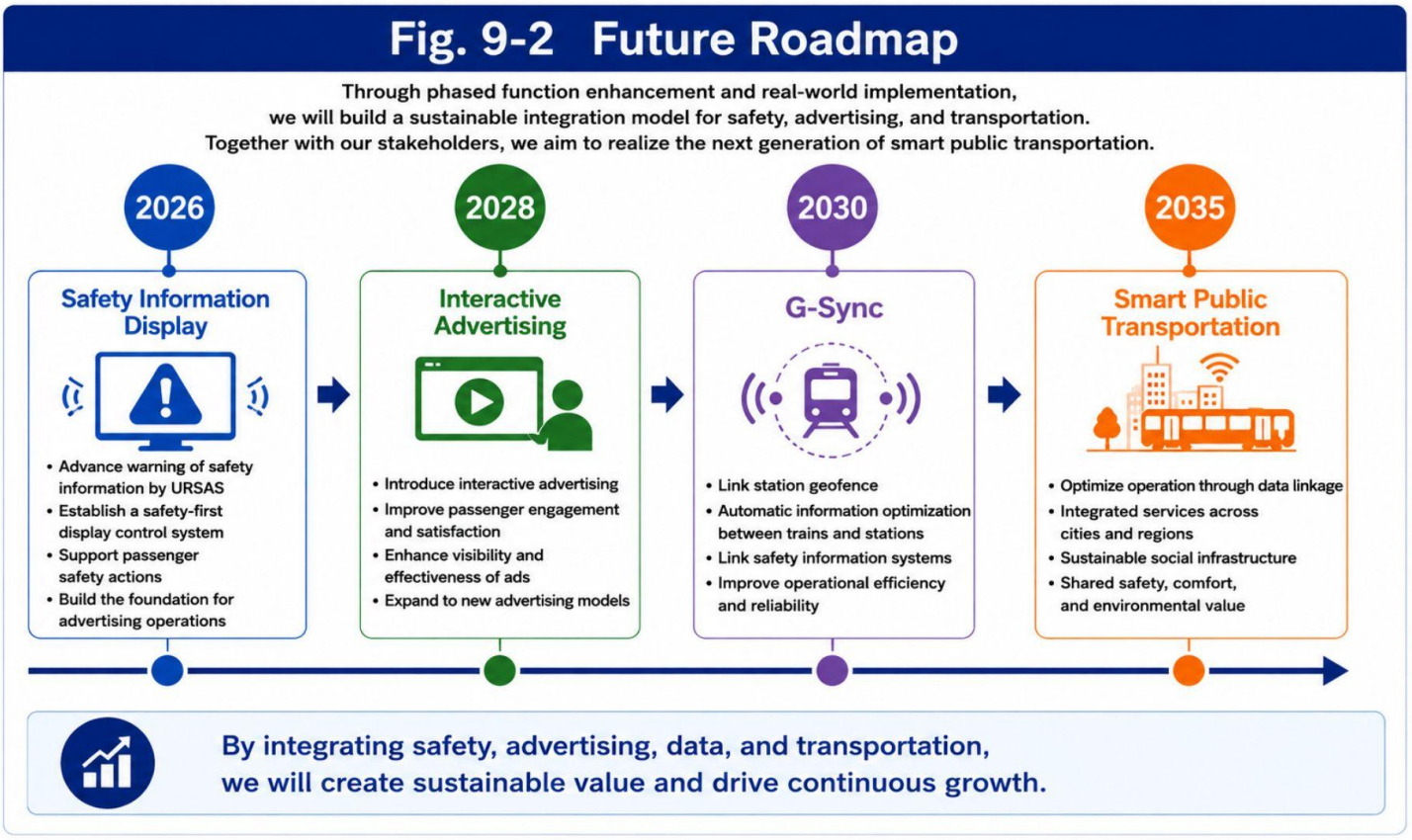
“a space for movement”

into:

“a space for experience.”

This represents a step toward a future model of public transportation in which:

Safety and Experience Value coexist.



Chapter 10: Universal Warning System

Universal Warning

10.1 Overview

Universal Warning is the integrated warning system within URSAS that communicates safety information to passengers.

The prediction results generated by:

- SCE
- DPE
- DCPE

as described in Chapters 5 through 7 are ultimately communicated to passengers through Universal Warning.

The purpose of URSAS is not simply:

to make predictions.

Its purpose is:

to create conditions in which passengers can take appropriate safety action.

Universal Warning is designed as the final output mechanism for achieving this objective.

Universal Warning distributes the notification content determined by the Tier Core to the appropriate notification devices.

If the EEW or Kill-Switch is activated, experience-oriented functions and advertising displays are suspended, while essential safety notifications continue with the highest priority.

10.2 Why Universal Warning Is Necessary

Even when the same information is provided, not every passenger can receive or perceive it in the same way.

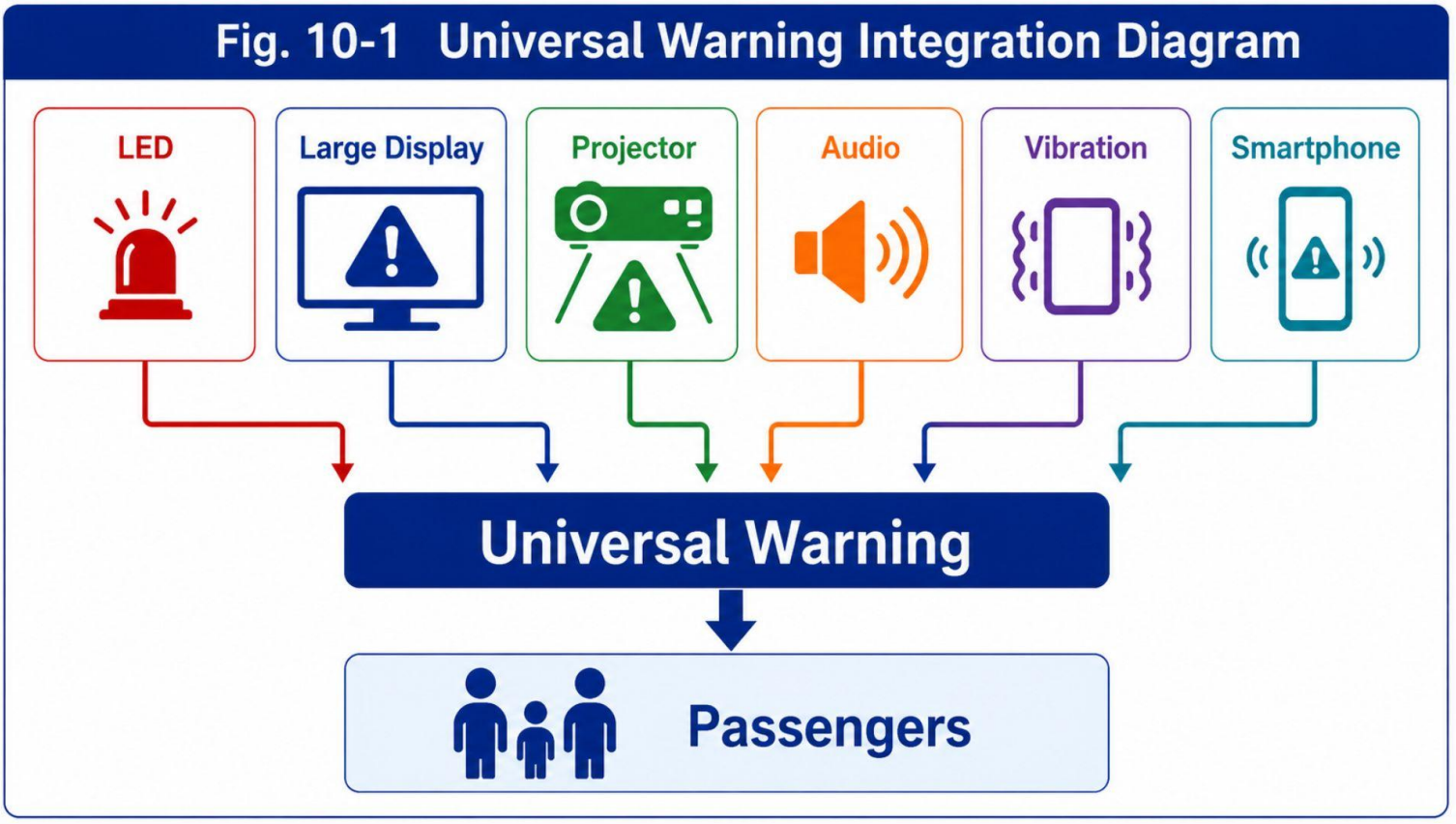
Passengers may include:

- people with visual impairments;
- people with hearing impairments;
- international travelers;
- older adults;
- passengers using smartphones; and
- passengers wearing headphones or earphones.

Passenger circumstances and communication needs therefore vary considerably.

For this reason, relying on a single notification method may not be sufficient.

URSAS aims to provide warnings that can be understood intuitively by as many passengers as possible by communicating through multiple sensory channels.



10.3 Multimodal Warning

URSAS adopts a:

Multimodal Warning

approach that combines:

Visual

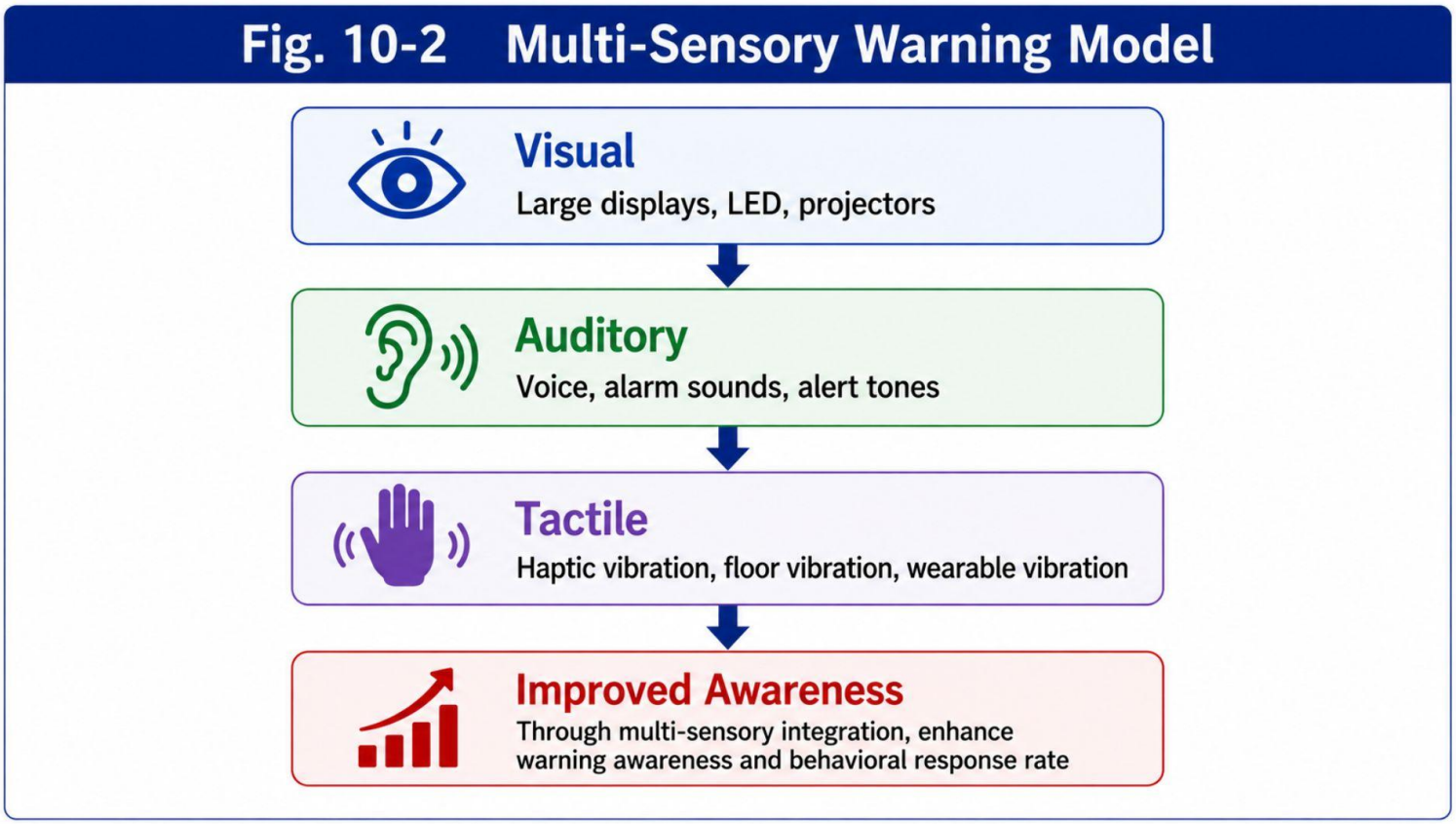
Auditory

Tactile

communication.

If a passenger does not perceive a warning through one channel, the same essential information may still be received through another.

This multimodal approach provides complementary communication pathways for safety information.



10.4 Visual Warnings

Visual warnings provide a primary means of communicating safety information to a broad range of passengers.

The main output devices include:

- large onboard displays;
- LED lines; and
- projectors.

In particular, the large onboard display functions as a core warning device within URSAS.

During normal operation, it provides passenger information and advertising content.

When a warning is activated, it immediately switches to the appropriate safety display.

10.5 Auditory Warnings

Auditory warnings are used to communicate information to passengers who may not be looking at the visual displays.

The main output methods include:

- onboard speakers; and
- brief audible notification tones.

To preserve a quiet and comfortable onboard environment, URSAS uses only the minimum necessary volume, duration, and number of audible notifications appropriate to the situation.

10.6 Tactile Warnings

Tactile warnings complement situations in which visual or auditory communication alone may not be sufficient.

The main output devices include:

- handrail / support-bar vibrators;
- floor vibration devices; and
- wearable devices.

In particular, tactile warnings delivered through handrails or other supports can provide intuitive alerts to passengers who are already holding them.

10.7 Smartphone and Wearable Integration

URSAS also supports integration with smartphones and wearable devices.

Using BLE (Bluetooth Low Energy), onboard warning information can be transmitted to compatible personal devices.

This provides an additional notification channel alongside visual, auditory, and tactile warnings.

10.8 Warning Control Based on Tier and Risk

Universal Warning changes its notification outputs according to the Tier and Risk Level determined by the URSAS control architecture.

Tier determines:

“Why is the warning being issued?”

Risk determines:

“How strongly should passengers be warned?”

Universal Warning uses the resulting decision from the Tier Core to deliver the appropriate combination of notifications for the situation.

Importantly, Universal Warning itself does not determine the Tier or Risk Level.

Its role is to deliver the notification content determined by the Tier Core through the appropriate output devices.

10.9 Relationship with the Human Behavior Transition Model

The purpose of URSAS is not simply to issue warnings.

It is to support appropriate changes in passenger behavior.

Universal Warning supports the progressive transition from:

Normal Operation

↓

Awareness

↓

Posture Preparation

↓

Safety Action

This is an important mechanism for implementing the Human Behavior Transition Model described in Chapter 4.

10.10 Significance of Universal Warning

URSAS is not simply a prediction system.

Nor is it simply a warning system.

What matters is that predicted and observed information is communicated in a form that passengers can recognize and understand, helping them take appropriate safety action.

Universal Warning is designed as the final output mechanism that connects URSAS safety decisions with passenger behavior.

Within URSAS, Universal Warning therefore serves as:

a bridge between people and technology.

Chapter 11: Safety Policy

11.1 Safety Policy

URSAS adopts a multilayer safety architecture that incorporates both external safety systems and internal safety mechanisms in order to place railway-system safety above all other functions.

The priority order for safety control is as follows:

EEW (Earthquake Early Warning)
↓
Kill-Switch (Final Safety Mechanism)
↓
Tier Core (Integrated Control During Normal Operation)
↓
Universal Warning (Passenger Safety Notification)

This structure ensures that the highest-priority safety instructions from the railway system take precedence, followed by URSAS internal safety mechanisms.

The Tier Core and Universal Warning operate only under normal conditions in which higher-priority emergency safety controls have not been activated.

URSAS is built on the design principle that:

Safety always comes first.

The system clearly defines the priority of each safety-related function and applies consistent priority control from external safety systems through to passenger notification.

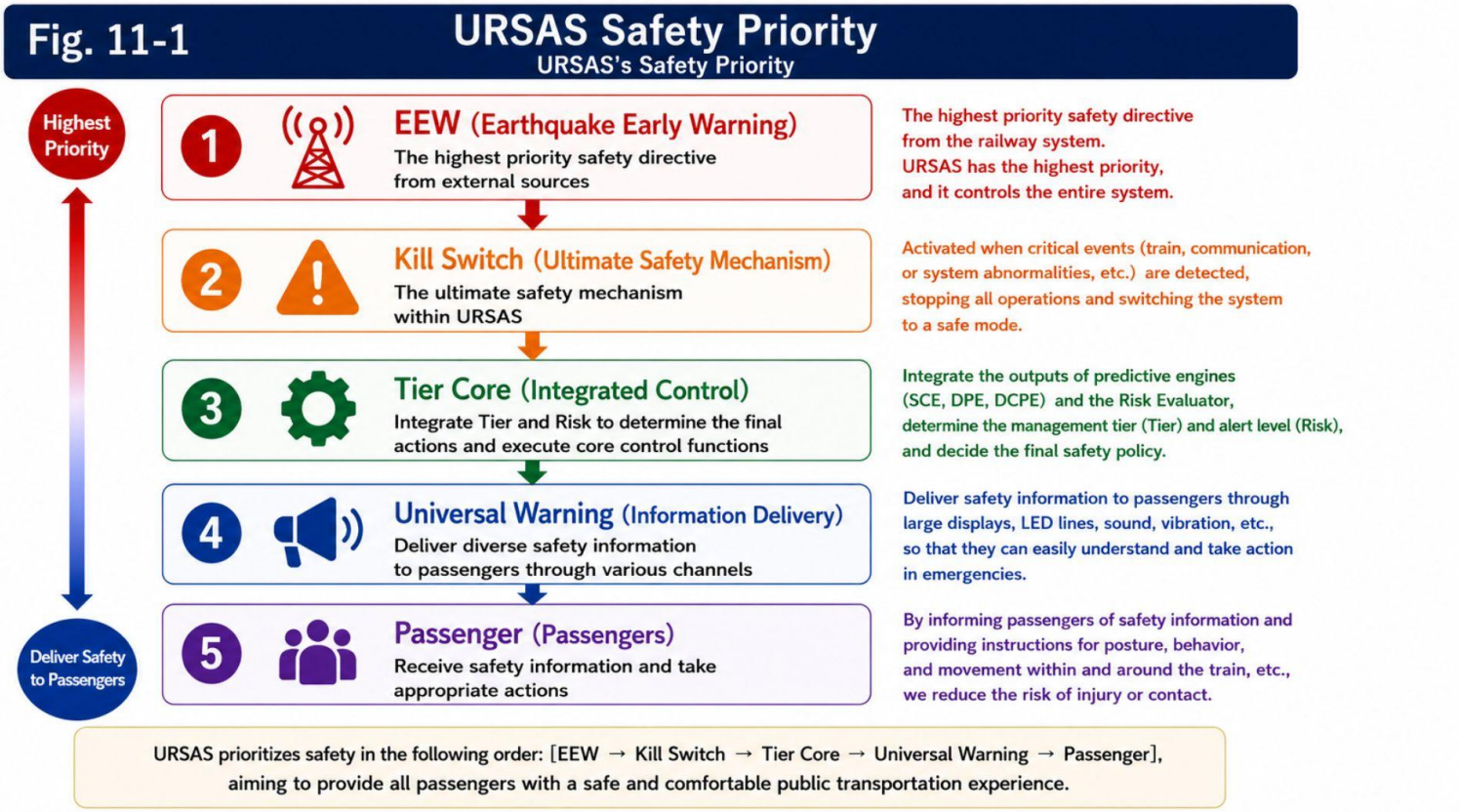


Figure 11-1 illustrates the priority structure of safety control within URSAS.

At the highest level is EEW (Earthquake Early Warning), followed by the Kill-Switch, which serves as the final internal safety mechanism within URSAS.

The Tier Core then integrates the outputs of the prediction engines and the Risk Evaluator, and safety information is delivered to passengers through Universal Warning.

This hierarchical structure enables consistent safety-priority control under all operating conditions.

11.2 Safety-First Principle

The fundamental safety principle of URSAS is to ensure that safety takes priority under all circumstances.

URSAS includes a variety of functions, such as:

- advertising;
- G-Sync;
- experiential effects; and
- passenger information services.

However, whenever a safety notification becomes necessary, all of these functions yield priority to safety-related functions.

This Safety-First Principle serves as a fundamental decision-making principle across the entire URSAS platform.

11.3 Safety Priority Order

URSAS applies the following order of control:

1. EEW — Earthquake Early Warning

↓

2. Kill-Switch
Final Safety Mechanism within URSAS

↓

3. Tier Core
Integration of Tier and Risk

↓

4. Universal Warning
Safety Notification

↓

5. Public Transportation Experience
G-Sync, Advertising, and Experiential Functions

This priority order does not change.

Safety-related functions always take precedence.

11.4 Safety Override

When a safety notification is required, URSAS automatically suspends or overrides experience-oriented functions.

For example, if the Tier Core determines that a safety notification is necessary while advertising is being displayed, the large onboard display immediately switches from advertising content to the appropriate safety information.

If the EEW or Kill-Switch is activated, normal Tier Core control is suspended and safety notification is given the highest priority.

This control principle is defined as:

Safety Override

11.5 What Is the Kill-Switch?

The Kill-Switch is the final internal safety mechanism within URSAS.

It causes the system to transition to a safe state in response to the highest-priority safety instruction from EEW or to serious events such as:

- vehicle abnormalities;
- communication failures; or
- system failures.

When the Kill-Switch is activated, normal Tier Core control is suspended and only safety-related notifications continue.

If an event occurs that exceeds the scope of normal Risk evaluation, all experience-oriented functions are suspended and the system transitions to:

Emergency Safety Mode

The Kill-Switch is designed as a control system independent of the normal Risk evaluation process.

It therefore functions as a final protective safety layer separate from Risk Levels 1 through 3.

11.6 Kill-Switch Activation Conditions

Examples of conditions that may activate the Kill-Switch include:

- Earthquake Early Warning (EEW);
- vehicle abnormalities;
- serious communication failures;
- system failures;
- emergency stops; and
- forced activation by the railway operator.

When one of these events occurs, the system transitions beyond normal warning control into Emergency Safety Mode.

11.7 System Behavior When the Kill-Switch Is Activated

When the Kill-Switch is activated, the following actions are executed:

- suspension of all advertising;
- suspension of all G-Sync functions;

- suspension of all experiential effects;
- safety information displayed on the large onboard displays;
- emergency indication through the LED lines;
- audible warnings;
- vibration-based warnings; and
- evacuation guidance displays.

These actions are executed automatically.

11.8 Integration with EEW — Earthquake Early Warning

URSAS is designed with integration with:

EEW (Earthquake Early Warning)

in mind.

When an EEW signal is received, the Kill-Switch is activated immediately.

This allows safety notifications to begin with the highest priority without waiting for normal Tier or Risk evaluation.

Earthquakes represent one of the most serious forms of unexpected events considered within the URSAS safety philosophy.

For this reason, EEW is treated as a special high-priority safety input.

11.9 Fail-Safe Design

URSAS adopts a:

Fail-Safe Design

in which the system transitions toward a safer state when a failure occurs.

If events such as:

- communication failure;
- power failure; or
- equipment failure

occur, experience-oriented functions are suspended while essential safety functions are maintained as far as the system configuration allows.

This design is intended to preserve safety priority even when system abnormalities occur.

11.10 Significance of the Safety Policy

URSAS is:

a safety system,

not:

an advertising system.

Although URSAS also seeks to enhance passenger experience value, safety is the foundation upon which all other functions depend.

The:

- Safety-First Principle
- Tier-Based Safety Experience Control
- Kill-Switch
- EEW Integration

are all designed according to this principle.

Within URSAS:

Safety is not simply one function among many.

Safety is the prerequisite for every other function.

Summary

URSAS is based on a safety-first design philosophy and applies control according to the priority structure:

EEW → Kill-Switch → Tier Core → Universal Warning

This multilayer safety architecture is designed to maintain consistency with railway-system safety requirements while appropriately coordinating:

- safety notifications;
- passenger experience functions; and
- revenue-generating functions.

Chapter 12: Expected Benefits of Implementation

12.1 Overview

URSAS is a public transportation support system developed to reduce risks associated with unexpected motion and sudden deceleration.

This chapter describes the benefits expected from the implementation of URSAS.

These benefits are organized into four main categories:

- Improved Safety
- Enhanced Passenger Experience
- Improved Operational Efficiency
- Enhanced Revenue Potential

12.2 Reduction of Fall Risk

On public transportation, unexpected motion or sudden deceleration can cause passengers to lose their balance and, in some cases, fall.

URSAS provides passengers with preparation time through progressive information sharing using:

- SCE
- DPE
- DCPE

This advance information is intended to encourage safety actions such as:

- holding onto a handrail or other support;
- stabilizing one’s footing; and
- adjusting one’s posture.

Through these actions, URSAS is expected to contribute to reducing the risk of passenger falls.

12.3 Reduction of Passenger Contact Incidents

Many unintended passenger-contact incidents inside vehicles can result from a loss of balance caused by unexpected motion.

By sharing information about upcoming motion in advance, URSAS aims to reduce:

- unintended contact between passengers;
- cascading loss of balance in crowded vehicles; and
- chain reactions in which multiple passengers lose stability.

These benefits may be particularly relevant during crowded commuting periods and other high-passenger-density conditions.

12.4 Increased Passenger Reassurance

The value provided by URSAS is not limited to physical safety.

Passengers may experience anxiety when they do not know what is about to happen.

By sharing advance information about the possibility of motion or deceleration, URSAS may contribute to improving passengers’ psychological sense of reassurance and preparedness.

This can be particularly valuable for passengers such as:

- older adults;
- tourists; and
- other passengers who may feel less confident in unfamiliar or crowded transportation environments.

12.5 Enhanced Passenger Experience

URSAS also aims to improve the value of travel time itself.

By utilizing functions such as:

- large onboard displays;
- tourism information;

- multilingual information services;
- G-Sync; and
- experiential advertising,

travel time can be transformed into an opportunity for information, local discovery, and immersive passenger experiences.

Through these functions, URSAS may contribute to increasing the overall attractiveness and perceived value of public transportation.

12.6 Improved Operational Efficiency

The introduction of large digital onboard displays can significantly reduce operational tasks associated with conventional printed advertising, including:

- printing;
- distribution;
- physical replacement; and
- disposal.

Remote updating of advertising and information content can also improve operational flexibility and efficiency.

12.7 Enhanced Advertising Value

Within URSAS, large displays installed primarily as safety infrastructure can also be used as advertising media during normal operation.

This enables forms of content such as:

- video advertising;
- location-linked advertising;
- tourism-related advertising; and
- experiential advertising.

In addition, coordinated advertising and visual effects can be presented across the vehicle environment, creating opportunities for advertising value beyond conventional printed onboard media.

12.8 Contribution to Implementation and Operating Costs

URSAS seeks to improve capital efficiency by integrating safety infrastructure and digital information infrastructure.

Large onboard displays installed for safety purposes can also generate advertising revenue during normal operation.

This revenue may help offset part of:

- implementation costs; and
- maintenance and operating costs.

This represents one of the characteristics of URSAS:

pursuing both improved safety and operational sustainability within the same platform.

12.9 The Value URSAS Seeks to Create

URSAS is not simply a warning system.

Nor is it simply an advertising system.

URSAS aims to establish a new model of public transportation that integrates:

Safety

Reassurance

Experience Value

Sustainability

By sharing information about unexpected motion in advance, URSAS seeks to help passengers travel with greater confidence and preparedness.

At the same time, by using safety infrastructure as a platform that can also support revenue-generating services during normal operation, URSAS may contribute to the development of a more sustainable implementation model.

URSAS therefore aims to establish a new standard for future public transportation by bringing together:

Safety and Operational Sustainability

without compromising the priority of passenger safety.

Chapter 13: RAMS Considerations and Validation Framework

13.1 Purpose and Scope

URSAS 5.4 is a conceptual public transportation safety platform intended to provide passengers with advance information about unexpected motion and sudden deceleration.

The purpose of this chapter is to identify the Reliability, Availability, Maintainability, and Safety (RAMS) considerations that should be addressed during future development and validation.

RAMS is introduced as an engineering evaluation framework. It does not constitute an additional prediction engine, warning-control function, or operational layer within URSAS.

The existing architecture, including SCE, DPE, DCPE, Tier Engine, Risk Evaluator, Tier Core, Universal Warning, and the safety-priority mechanisms described in Chapter 11, remains unchanged.

This chapter identifies evaluation objectives and areas requiring further investigation. It does not establish that URSAS has achieved a specified level of RAMS performance or compliance with railway safety standards.

13.2 Reliability — Consistency of Warning Performance

Reliability concerns the ability of URSAS to perform its intended functions consistently under defined operating conditions.

For URSAS, reliability should be evaluated across the three prediction and information-sharing technologies:

SCE — Statistical Complement Engine

Evaluation should consider the accuracy and relevance of accumulated Observed Motion data, the statistical treatment of historical information, and the updating of route-related motion-risk information.

DPE — Deceleration Prediction Engine

Evaluation should consider prediction accuracy under different operating conditions, including changes in speed, position, deceleration patterns, and the quality of available operational information.

DCPE — Deterministic Chain Prediction Engine

Evaluation should consider the accuracy of motion-point identification, ETA calculation, preceding Reference Vehicle selection through the V-Referencer, and information sharing between vehicles.

Across all three technologies, future testing should examine false warnings, missed warnings, data uncertainty, processing delays, and communication latency.

The objective is to determine the conditions under which URSAS can provide sufficiently reliable advance information.

13.3 Availability — Access to Warning Functions

Availability concerns whether the intended URSAS functions can be accessed and used when required.

Future evaluation should examine the effects of:

- Sensor unavailability or degraded measurements
- Communication interruption between vehicles
- Temporary loss of external information
- Onboard processing or device failures
- Partial unavailability of passenger notification devices

The three-tier architecture provides different sources of information, but the presence of multiple prediction methods does not automatically establish system redundancy or guaranteed availability.

The ability of SCE to provide historical route-risk information may complement the limitations of DPE and DCPE. However, the effectiveness of this complementary relationship requires validation.

When reliable warning information is unavailable, the system should not present uncertain predictions as confirmed events.

The required behavior under degraded or unavailable conditions must be defined and tested before operational deployment.

13.4 Maintainability — Sustainable System Operation

Maintainability concerns the ability to inspect, diagnose, repair, update, and restore URSAS functions throughout their operational life.

URSAS includes onboard sensors, processing devices, communication interfaces, prediction functions, and multimodal warning devices.

Future maintainability requirements should address:

- Fault detection and diagnostic information
- Inspection and maintenance procedures
- Software and configuration management
- Sensor calibration and verification
- Replacement of onboard components
- Recording and analysis of system failures
- Verification following repairs or software updates

Maintenance procedures should minimize unnecessary disruption to existing railway operations.

The feasibility and maintenance burden of the proposed system must be evaluated in cooperation with railway operators and relevant engineering specialists.

13.5 Safety — Prevention of System-Induced Hazards

Safety evaluation must consider not only the potential benefits of advance warnings but also hazards that URSAS itself could introduce.

Examples of potential hazards include:

- False warnings that encourage unnecessary passenger movement
- Missed warnings that create inappropriate expectations of protection
- Incorrect ETA information
- Excessive warning frequency and warning fatigue
- Conflicting information from multiple notification channels
- Warning methods that distract, startle, or confuse passengers
- Failure to provide accessible information to passengers with different needs
- Unintended interactions with existing railway systems

A structured hazard analysis should identify possible causes, consequences, and appropriate risk-control measures.

URSAS is not intended to control train movement, signaling, or braking systems.

Its proposed safety function is to support passenger awareness and preparation through advance information.

The safety of the passenger warning function must therefore be evaluated within its defined operational boundaries.

13.6 Relationship to the Existing Safety Policy

The safety-priority architecture defined in Chapter 11 remains the governing principle of URSAS.

The priority order is:

EEW (Earthquake Early Warning)

↓

Kill-Switch (Final Safety Mechanism)

↓

Tier Core (Integrated Control During Normal Operation)

↓

Universal Warning (Passenger Safety Notification)

Public Transportation Experience functions remain subordinate to safety-related functions.

The Kill-Switch is an internal URSAS safety mechanism and is not a device for stopping or controlling the train.

RAMS evaluation should examine whether the intended safety-priority behavior can be maintained during normal operation, abnormal conditions, communication failures, and relevant emergency scenarios.

This chapter does not modify the responsibilities of the Tier Engine, Risk Evaluator, Tier Core, or Universal Warning.

13.7 Human Factors and Passenger Response

The intended value of URSAS depends on how passengers interpret and respond to advance warnings.

Generating a warning does not, by itself, demonstrate that the warning improves passenger safety.

Future human-factors evaluation should investigate:

- The amount of preparation time required for useful safety action
- Passenger recognition and understanding of Risk Levels 1–3
- Response differences between standing and seated passengers
- The effectiveness of visual, audible, and tactile notifications
- Passenger responses under crowded conditions
- Accessibility for passengers with different physical and sensory needs
- The potential effects of repeated warnings on attention and trust

The Human Behavior Transition Model described in Chapter 4 should be treated as a conceptual model until its assumptions and expected benefits have been experimentally evaluated.

13.8 Proposed Validation Framework

Future development should use a staged validation process.

Stage 1 — Requirements Definition

Define the intended operating conditions, system boundaries, warning objectives, safety requirements, and measurable performance criteria.

Stage 2 — Simulation and Algorithm Evaluation

Evaluate SCE, DPE, and DCPE using representative data and simulated operating scenarios.

Assess prediction accuracy, ETA estimation, warning timing, and sensitivity to data uncertainty.

Stage 3 — Prototype and Integration Testing

Develop and test representative onboard processing, communication, and warning functions.

Verify information flow, component interactions, and safety-priority behavior.

Stage 4 — Human-Factors Evaluation

Conduct controlled studies to assess passenger understanding, response time, safety actions, and unintended behavioral effects.

Stage 5 — Railway Environment Evaluation

Subject to appropriate authorization and safety controls, evaluate the system under representative railway operating conditions.

Stage 6 — Independent Technical and Safety Review

Review the evidence, identified hazards, residual risks, operational limitations, and readiness for any further development or deployment consideration.

These stages are proposed future activities and do not represent tests already completed.

13.9 Relationship to Railway RAMS Standards

Railway RAMS standards, including the EN 50126 series, provide an established framework for considering reliability, availability, maintainability, and safety throughout the railway system lifecycle.

Future URSAS development may use relevant railway standards to support:

- Requirements definition
- Hazard identification and risk assessment
- Lifecycle planning
- Verification and validation
- Documentation and traceability
- Safety assurance activities

The applicability of specific standards, requirements, and assessment procedures must be determined according to the intended system scope and deployment environment.

URSAS 5.4 has not demonstrated compliance with EN 50126 or any other railway safety standard.

No railway safety certification or formal safety approval is claimed.

13.10 Development Status and Future Direction

URSAS 5.4 remains a conceptual research and development proposal.

Its prediction performance, passenger safety benefits, operational availability, maintainability, and overall safety characteristics have not yet been established through comprehensive validation.

The introduction of RAMS considerations does not change the fundamental URSAS philosophy:

To create time for passengers to prepare for unexpected motion.

Instead, RAMS provides a structured approach for examining whether this philosophy can be translated into a reliable, maintainable, and safe passenger-information system.

Future progress will depend on engineering evaluation, experimental evidence, human-factors research, and constructive review by railway professionals and safety specialists.

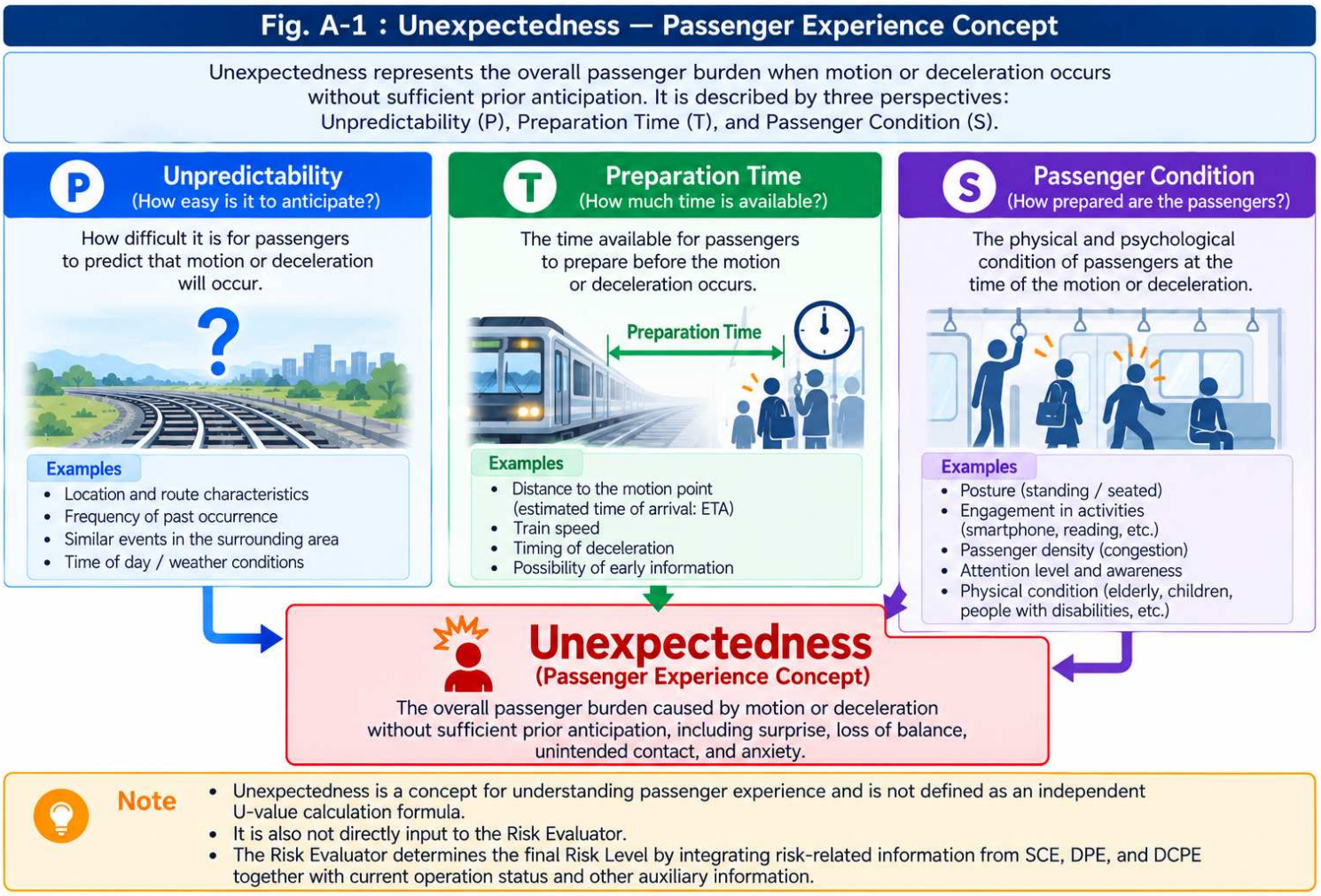
The ultimate objective is not simply to generate advance warnings, but to establish whether those warnings can provide measurable passenger safety benefits without introducing unacceptable risks.

Appendix

A. Glossary

Term	Definition
URSAS	Abbreviation for Universal Rail Safety Alert System. A public transportation safety support system designed to share advance information about unexpected motion and sudden deceleration.
Tier	A hierarchical classification representing the reason for a warning. URSAS uses Tier 1 through Tier 3.
Risk	An evaluation index representing warning intensity. URSAS uses Risk Levels 1 through 3.
Tier Core	The central control function that integrates Tier, Risk, and Kill-Switch status to determine the final safety policy.
Universal Warning	An integrated warning system that communicates safety information to passengers through visual, auditory, and tactile channels.
Public Transportation Experience	The URSAS experience-design philosophy that seeks to enhance the value of travel time while maintaining safety as its foundation.
Safety Cue	Advance precautionary information that gently informs passengers of the possibility of upcoming motion or deceleration.
Track Condition	A general term referring to track conditions, including track settlement, track irregularities, maintenance status, and related route characteristics.
Station Geofence	A location-based management function that defines stable and predictable acceleration and deceleration zones around stations.
G-Sync	Geo-Synchronized Experience. An immersive experience system that synchronizes experiential effects with acceleration and deceleration within designated station geofences.
Kill-Switch	The final internal safety mechanism within URSAS. In response to EEW or serious events such as vehicle, communication, or system abnormalities, it suspends normal Tier Core control and transitions the system to a safety-priority mode.

Vehicle Motion	A general term covering vehicle-body movement during train operation, including lateral motion, vertical motion, vibration, and rapid deceleration.
Observed Motion	Motion that has actually occurred and has been detected by onboard sensors such as IMUs and accelerometers. DCPE uses Observed Motion to calculate the ETA to a motion point.
Motion Information	A general term for analyzed information relating to vehicle motion, including Observed Motion, motion history, motion intensity, and ETA.
Vibration	Small or periodic physical motion transmitted through the vehicle body, floor, handrails, or other structures. In URSAS, the term is also used for tactile notification devices such as floor vibration devices.
Lateral Motion	Side-to-side vehicle movement that may contribute to instability or unintended passenger contact, particularly among standing passengers.
Rapid Deceleration	A condition in which a train decelerates at a greater rate than during ordinary operation. DPE treats the possibility of such deceleration as a prediction target.
Deceleration	A general operating condition in which train speed decreases, including ordinary station stopping. It is distinguished from Rapid Deceleration.
Impact	A short-duration force or load experienced by passengers as a result of a rapid change in acceleration, such as that associated with rail joints or other localized inputs.
Unexpectedness	The overall passenger burden caused when motion or deceleration occurs without sufficient prior anticipation, including surprise, loss of balance, unintended contact, and anxiety.



Note: Figure A-1 presents Unpredictability (P), Preparation Time (T), and Passenger Condition (S) as conceptual perspectives for understanding Unexpectedness from the passenger-experience perspective.

These concepts are intended to describe the passenger experience and are not defined as an independent U-value calculation formula or as a direct input model to the Risk Evaluator.

A-1. Terminology Policy

Terminology Usage in This White Paper

To ensure consistency throughout this white paper, the following terminology rules are applied.

Term	Usage
Vehicle Motion	Used as the general term for vehicle-body behavior occurring during train operation.
Observed Motion	Used for motion actually detected by onboard sensors. In DCPE, Observed Motion is used to calculate the ETA to a motion point.
Motion Information	Used as the general term for analyzed motion-related information, including motion history, Observed Motion, motion intensity, and ETA.

Vibration	Used for physical vibration transmitted through the vehicle body, floor, handrails, or other structures, and for tactile notification functions such as floor vibration devices.
Lateral Motion	Used to describe side-to-side vehicle motion, particularly when discussing instability or contact risk for standing passengers.
Deceleration	Used for the general condition in which train speed decreases, including ordinary station stopping.
Rapid Deceleration	Used for deceleration greater than ordinary operating levels and as an event predicted by DPE.
Impact	Used when referring to short-duration forces resulting from sudden changes in acceleration.

A-2. Terminology Usage Policy

For general explanations, this white paper uses Vehicle Motion as the basic umbrella term to make the system easier to understand.

When describing the processing performed by each URSAS engine, the terminology is differentiated as follows:

- SCE: Uses motion history and analytical results related to route characteristics and Track Condition.
- DPE: Predicts the possibility of Rapid Deceleration.
- DCPE: Uses Observed Motion detected by a preceding vehicle to calculate the ETA to the Motion Point and provides the resulting information for the Following Vehicle immediately behind the Warning-Responsible Vehicle.
- Risk Evaluator: Integrates the analyzed output information produced by the prediction engines and determines the appropriate Risk Level.

A-3. Symbols and Abbreviations

The following abbreviations are used consistently throughout this white paper.

Abbreviation / Term	Meaning
ETA	Estimated Time of Arrival at the Motion Point; in URSAS, this refers to the estimated time remaining until a vehicle reaches the motion point.
Track Condition	Track condition and route characteristics
Tier	Warning Reason — Why to Warn

Risk	Warning Intensity — How Strongly to Warn
EEW	Earthquake Early Warning

B. List of Abbreviations

Abbreviation	Full Name
URSAS	Universal Rail Safety Alert System
SCE	Statistical Complement Engine
DPE	Deceleration Prediction Engine
DCPE	Deterministic Chain Prediction Engine
MCU	Microcontroller Unit
IMU	Inertial Measurement Unit
GPS	Global Positioning System
BLE	Bluetooth Low Energy
EEW	Earthquake Early Warning
RAMS	Reliability, Availability, Maintainability and Safety
ETA	Estimated Time of Arrival at the Motion Point
LED	Light-Emitting Diode
UI	User Interface
V-Referencer	Variable Referencer
G-Sync	Geo-Synchronized Experience

C. Symbols

Symbol / Term	Meaning
Δ	Reference-vehicle offset (number of vehicles ahead) selected by the V-Referencer

ETA	Estimated time remaining until a vehicle reaches the motion point
Tier 1	Route-risk sharing by SCE
Tier 2	Deceleration-risk sharing by DPE
Tier 3	Observed-motion sharing by DCPE
Risk 1	Attention — Awareness Phase
Risk 2	Caution — Preparation Phase
Risk 3	High Alert — Safety Action Phase

D. References and Technical Background

This white paper was developed with reference to general concepts and knowledge from the following technical fields:

- railway engineering;
- ergonomics;
- human factors engineering;
- public transportation safety engineering;
- risk management;
- railway RAMS (Reliability, Availability, Maintainability and Safety);
- railway safety assurance and lifecycle management;
- fail-safe design;
- universal design;
- ITS (Intelligent Transport Systems);
- railway traffic and operation management systems;
- railway vehicle dynamics and vibration engineering; and
- Earthquake Early Warning (EEW) technologies.

Relevant Standards and Frameworks

The EN 50126 series provides a reference framework for railway RAMS lifecycle management, requirements definition, risk assessment, verification, and validation.

URSAS 5.4 has not been assessed or certified for compliance with EN 50126 or any other railway safety standard. The applicability of specific standards and requirements remains subject to future technical evaluation.

The URSAS architecture and the concepts described in this white paper, including:

- Tier-Based Safety Experience Control
- the three-layer prediction architecture consisting of SCE, DPE, and DCPE
- Universal Warning
- G-Sync

- Station Geofence
- V-Referencer
- Human Behavior Transition Model

are original concepts proposed by the author.

This document is a technical white paper describing a concept currently at the research and development stage.

Implementation methods and system specifications may be revised as development progresses.

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URSAS

Universal Rail Safety Alert System

Version 5.4

We cannot eliminate motion.

But we can become aware of it a little earlier.

Those few seconds can protect people.

URSAS